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Department of Mechanical Engineering
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ELEMENTS OF MECHANICAL ENGINEERING

UNIT- III
TURBINES, STEAM & ITS PROPERTIES

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UNIT- III (5 hours)

Steam and its properties, property charts.

Steam turbines: classification and working

Hydraulic Turbines: Classification of hydraulic turbines, working of Pelton, Francis and Kaplan turbines; comparison between impulse and reaction turbines.

Gas Turbines: Working of Gas Turbines (Brayton cycle).

Refrigeration: Refrigeration effect, working principle of Vapour Compression refrigeration systems, ton of refrigeration, COP, refrigerants and their properties.

Prime Mover

A Prime mover is a self moving device which converts the available natural source of energy in to mechanical energy of motion to drive the other machines.

The various types of prime movers which convert heat energy produced by the combustion of fuels in to mechanical energy are ***steam engines, steam turbines, gas turbines, internal combustion engines***.

STEAM TURBINES

Steam turbine: it is defined as a prime mover in which heat energy of the steam is transformed in to mechanical energy directly in the form of rotary motion.

The basic principle of operation of steam turbine involves the generation of high pressure steam jet by combustion of fuels and letting the high pressure and temperature steam to expand in a nozzle to convert its kinetic energy in to mechanical work on rotor blades.

Steam turbine is mainly used as an ideal prime mover to drive the electric generators to generate the electric power in thermal power plants. It is also used in ship propellers, pumps, compressors, textile & sugar industry machinery, etc.

Classifications of Steam Turbines

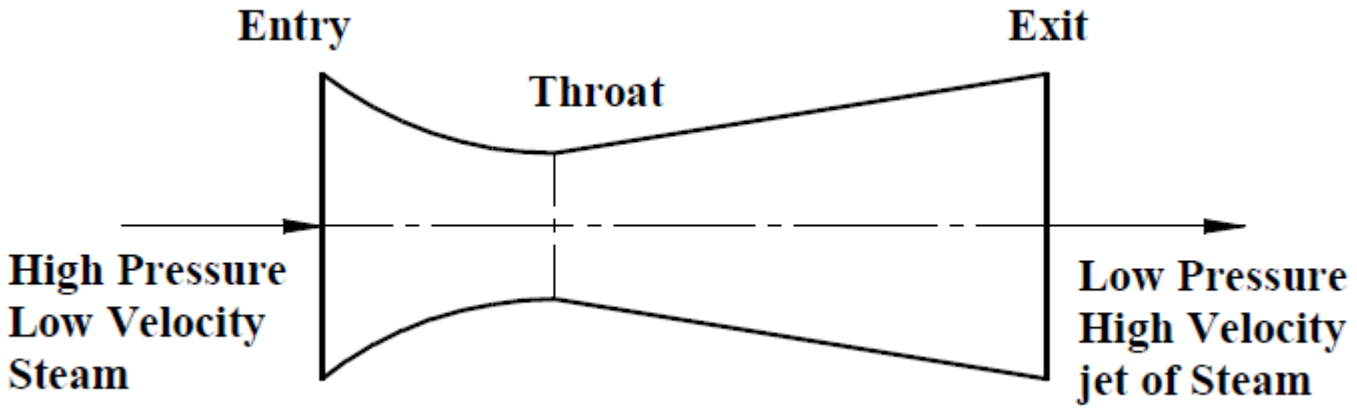
Steam turbines are mainly classified into two groups.

- ***Impulse steam turbine***
- ***Reaction steam turbine***
- In an impulse turbine the steam expands in the nozzles and its pressure does not alter as it moves over the blades.
- where as in a reaction turbine, the steam expands continuously as it passes over the blades and thus, there is a gradual fall in the pressure during expansion

Impulse steam turbine:

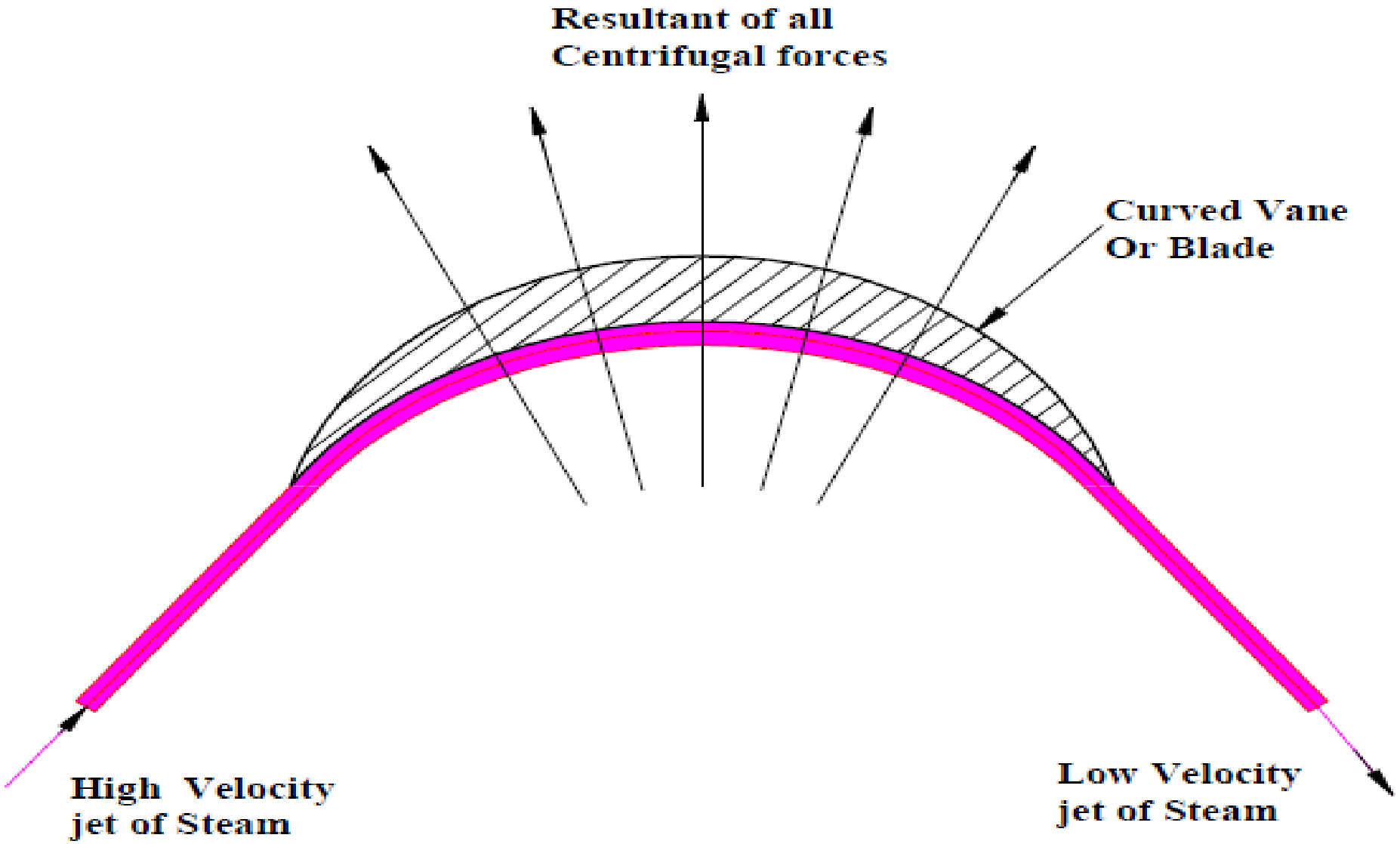
- In impulse turbines, the steam is expanded from its initial high pressure to a lower pressure before it is delivered to the moving blades on the rotor.
- The pressure of the steam over the blades will be low and the velocity of the steam continuously decreases as it glides over the blades where in the kinetic energy of the steam gets converted in to mechanical energy of rotation.

Expansion in Nozzle:



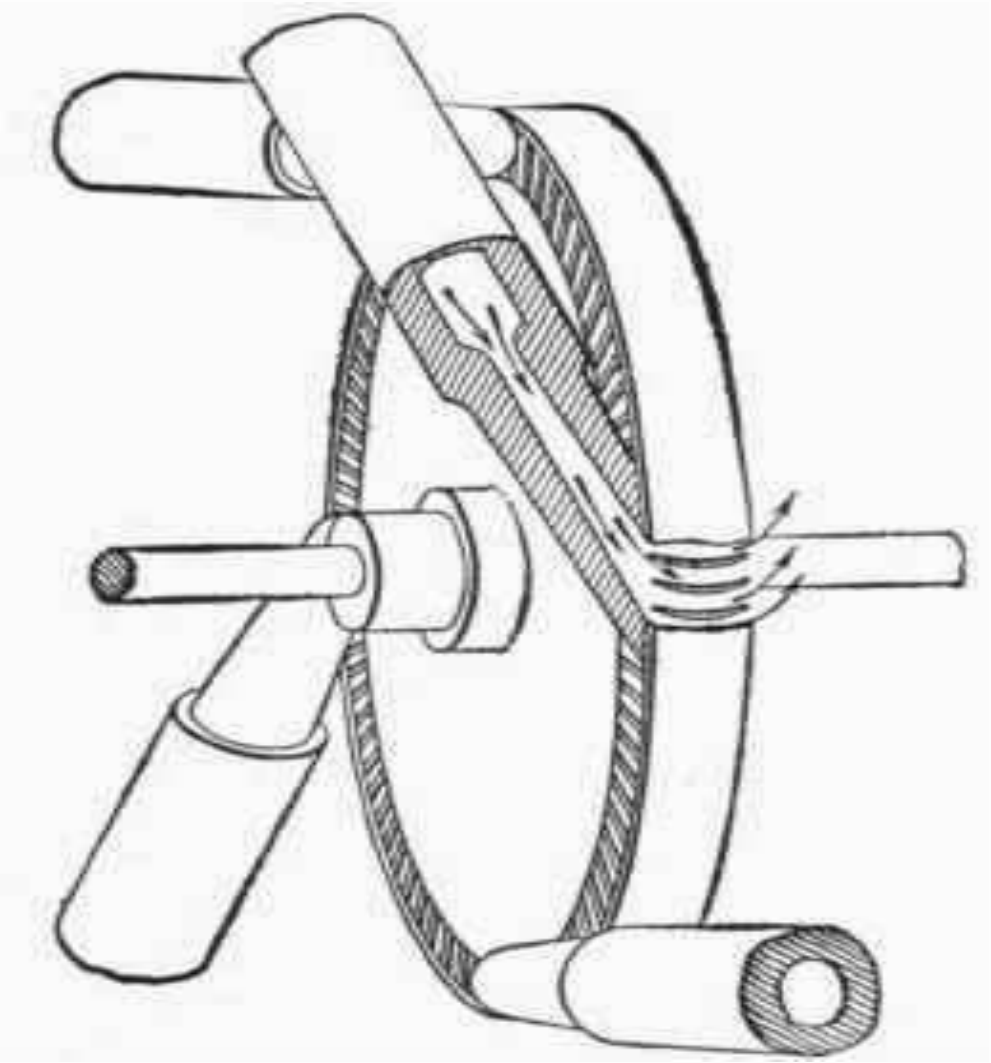
A high velocity jet of steam is produced by expanding a high pressure steam in a convergent –divergent nozzle as shown in fig. Steam at high pressure and low velocity enters the nozzle and as it passes between the entry and throat, it expands to a low pressure & high velocity and comes out through the divergent portion.

Propelling force on Impulse Turbine:



Impulse turbine Rotor

Impulse turbine Rotor

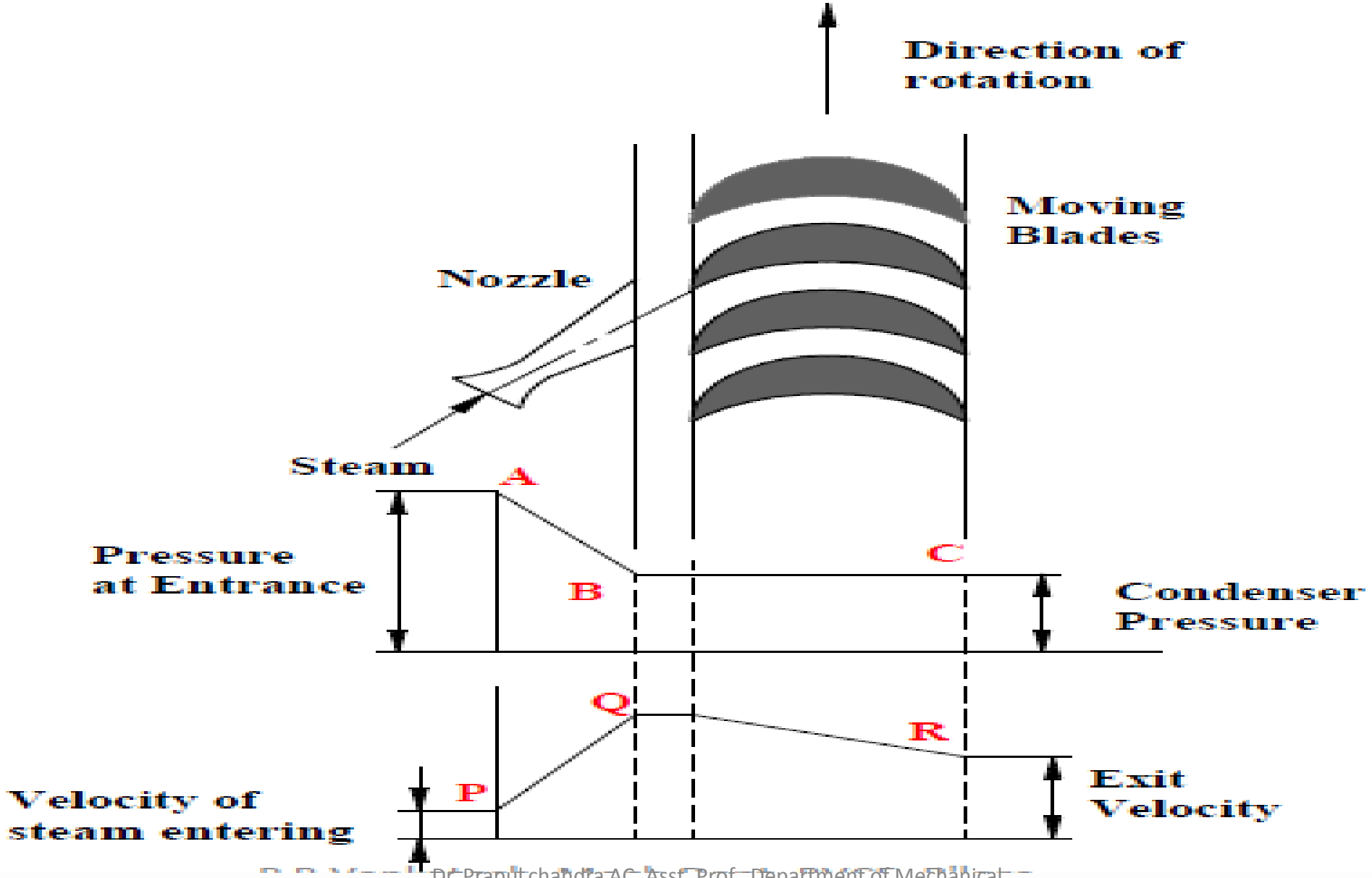




Impulse-Momentum principle

- The high velocity jet of steam coming out of the nozzle is made to glide over a curved vane (***blade***) which makes the jet to get deflected very nearly in the circumferential direction.
- This causes the steam jet to under go a change in direction which gives rise to ***change in momentum*** and hence a ***force*** which will be centrifugal in nature.
- The centrifugal force causes the ***rotor*** which has the blades on its periphery.

Impulse turbine



Impulse turbine

The fig shows the diagrammatic representation of an impulse turbine.

The lower portion shows the nozzle and the blade & the top portion shows the corresponding variation of pressure and velocity of the steam as it flows through the nozzle and over the blades.

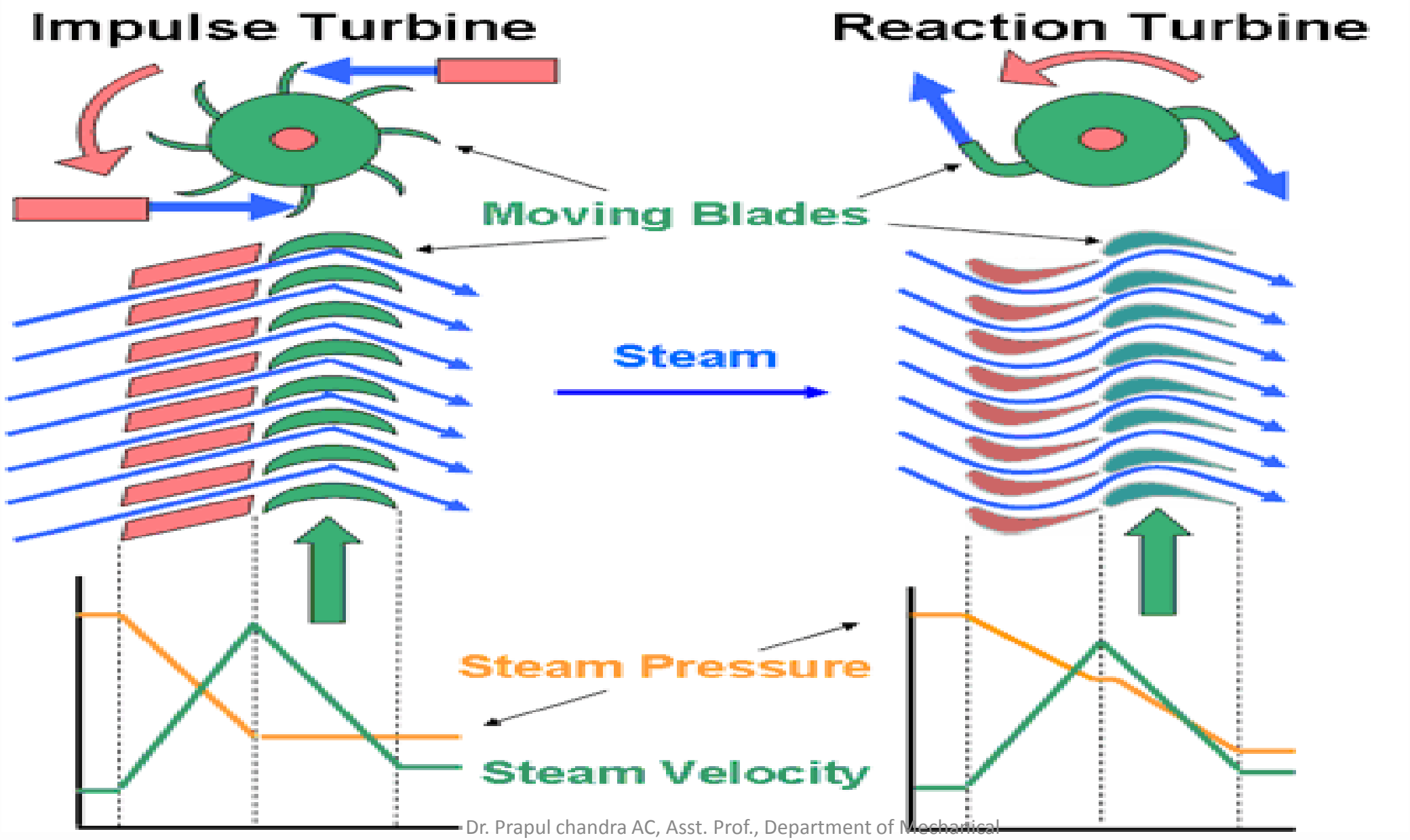
Since, the expansion of the steam takes place in the nozzle, the pressure drop is represented by the curve AB, the corresponding rise in velocity in the nozzle is represented by the curve PQ.

As there is no change in the pressure of steam that is passing over the blade, this flow is represented by the horizontal line BC.

As the blades absorb the kinetic energy of the steam as it flows over it, the velocity decreases. This is represented by the curve QR.

The principal example of this turbine is ***De-Lavalturbine.***

Impulse turbine



Reaction turbine

In this type of turbine, the high pressure steam does not initially expand in the nozzle as in case of impulse turbine, but instead directly passes on the moving blades whose shapes are designed in such a way that the steam flowing between the blades will be subjected to the nozzle effect.

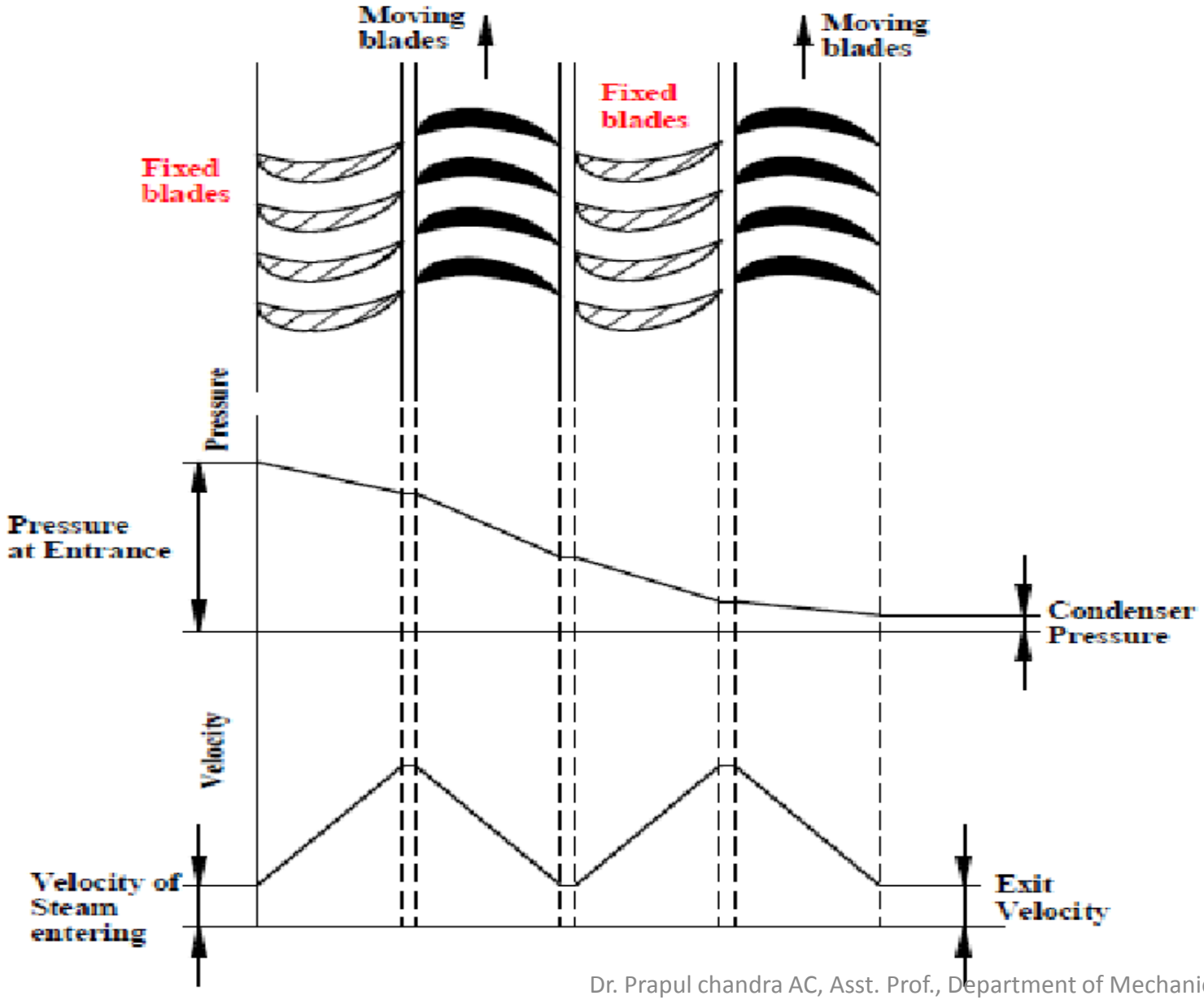
•Hence, the pressure of the steam drops continuously as it flows over the blades causing simultaneous increase in the velocity of the steam.

It is also called as impulse-reaction turbine. It consists of a number of rows of moving blades fitted on the different rotors keyed to the turbine shaft with alternate rings of fixed blades rigidly fixed to the casing of the turbine.

Reaction turbine

- Both the fixed and moving blades are designed in the shape of the nozzles.
- Therefore, the expansion of the steam takes place both in the fixed and moving blades.
- The fixed blade ring between the two moving blade rotors enables to deflect and glide the steam to enter from one row of moving blades to the next row

Reaction turbine

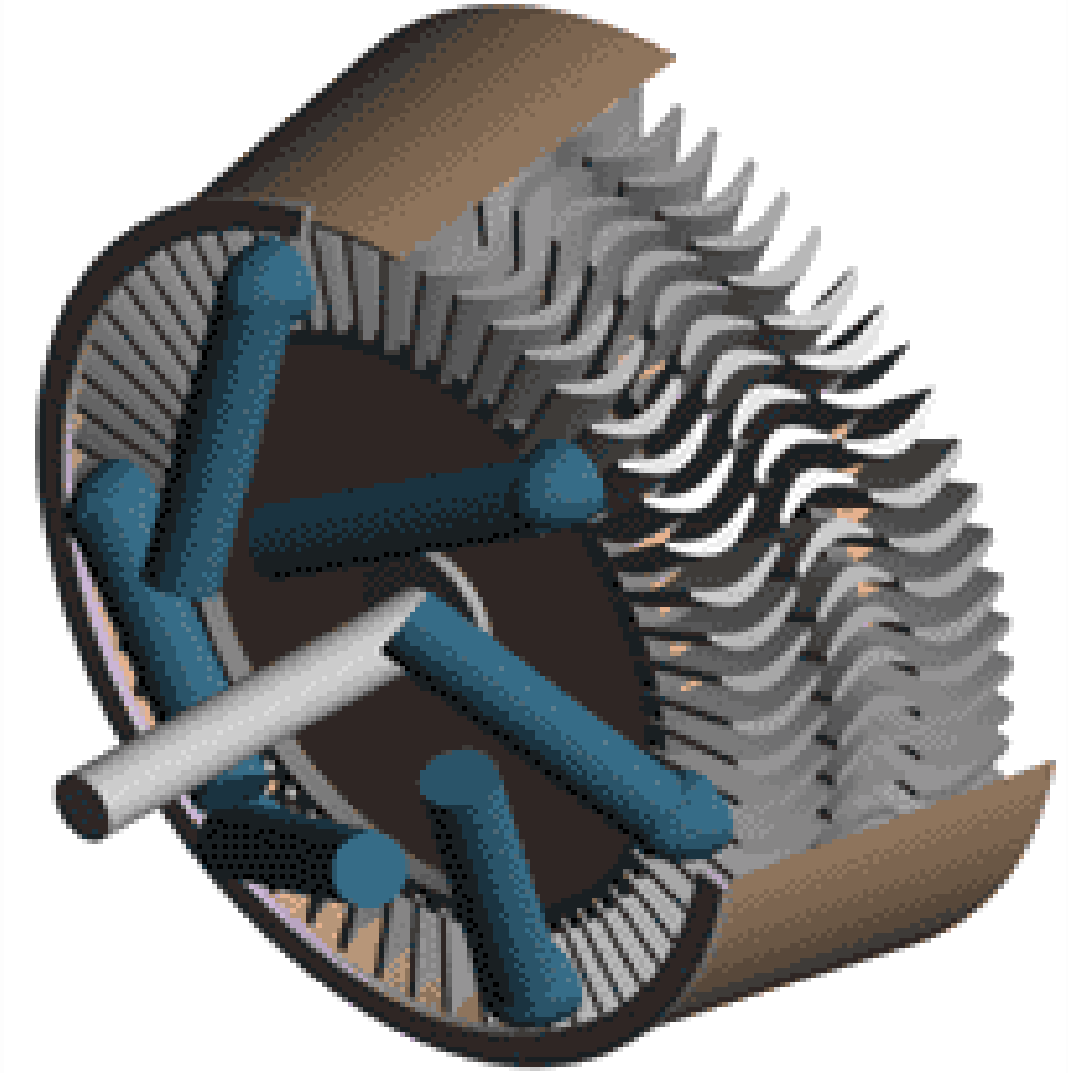


Reaction turbine

The high pressure steam passing in the first row of fixed blades under goes a small drop in pressure causing the increase in the velocity of the steam.

- It then enters the first row of moving blades where it suffers further drop in pressure and the velocity energy is converted in to the mechanical energy of rotation of the rotor.
- Thus, the velocity of the steam decreases.This continues in the further rows of moving and fixed blades till the pressure of the steam is almost completely reduced.
- The changes in the pressure and velocity of the steam as it flows over the moving and fixed blades are as shown in the fig

Reaction turbine



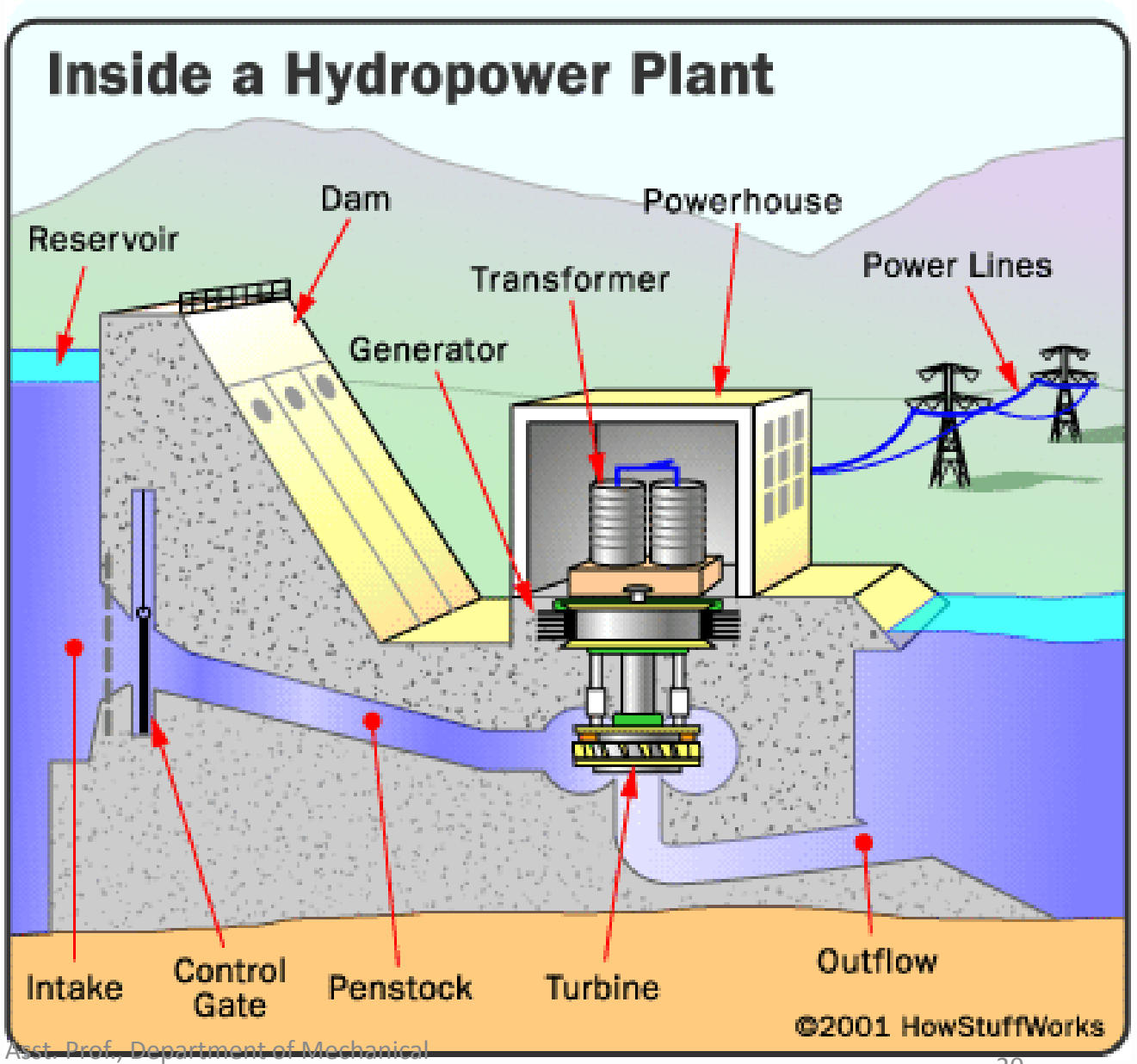
Comparison Between Impulse and Reaction Steam turbines

Impulse Steam Turbine	Reaction Steam turbines
The steam completely expands from a high pressure to a low pressure in the nozzle before it enters on the moving blades.	The high pressure steam continuously expands successively in both the fixed & moving blades.
Turbine blades have symmetrical profile.	Turbine blades have aerofoil profile.
The pressure on both ends of the moving blade is same.	The pressure on two ends of the moving blades are different
Since there is a large drop in pressure in the nozzle, the steam speed as well as the rotor speed are high.	Due to smaller pressure drop over both fixed and moving blades, both the steam speed and rotor speed are relatively low.
Because of the larger pressure drop in the nozzle and less no. of stages, size of the impulse turbine for the same power output is comparatively small.	Because of smaller pressure drops in every stage and more no. of stages, the size of the reaction turbine for the same power output is large.
occupies less space per unit power.	Occupies more space per unit power.
Suitable for small power generation prime movers	Suitable for medium and high power generation prime movers.
Due to high rotor speed, the compounding is required to reduce the speed	Since, the speeds are relatively less, the compounding is not required.
Ex: De Laval Turbine	***Ex: Parson's Turbine***

HYDRAULIC TURBINES

A *hydraulic machine* that converts the hydraulic energy into mechanical energy is called a ***turbine***, while the machine that converts the mechanical energy into hydraulic energy is called a ***pump***.

A water turbine is a rotary device that takes energy from moving water



Classification of hydraulic turbines

The classifications of water turbines are as follows:

According to the type of energy inlet:

Impulse or velocity turbine.- **Kinetic energy- Ex: Pelton wheel**

Reaction or pressure turbine – **both pressure energy and kinetic energy- Ex: Kaplan and Francis**

According to the direction of flow of water through runner:

Tangential flow turbine - **Ex: Pelton wheel**

Radial flow turbine – **Ex: Francis**

Axial flow turbine – **Ex: Kaplan**

Mixed flow turbine – **Ex: Modern Francis turbine**

According to the available water head at inlet:

High head turbine – **Ex: Pelton wheel** (several 100 – few 1000m)

Medium head turbine and – **Ex: Francis turbine** (50- 400m)

Low head turbine – **Ex: Kaplan (<50m)**

IMPULSE TURBINES

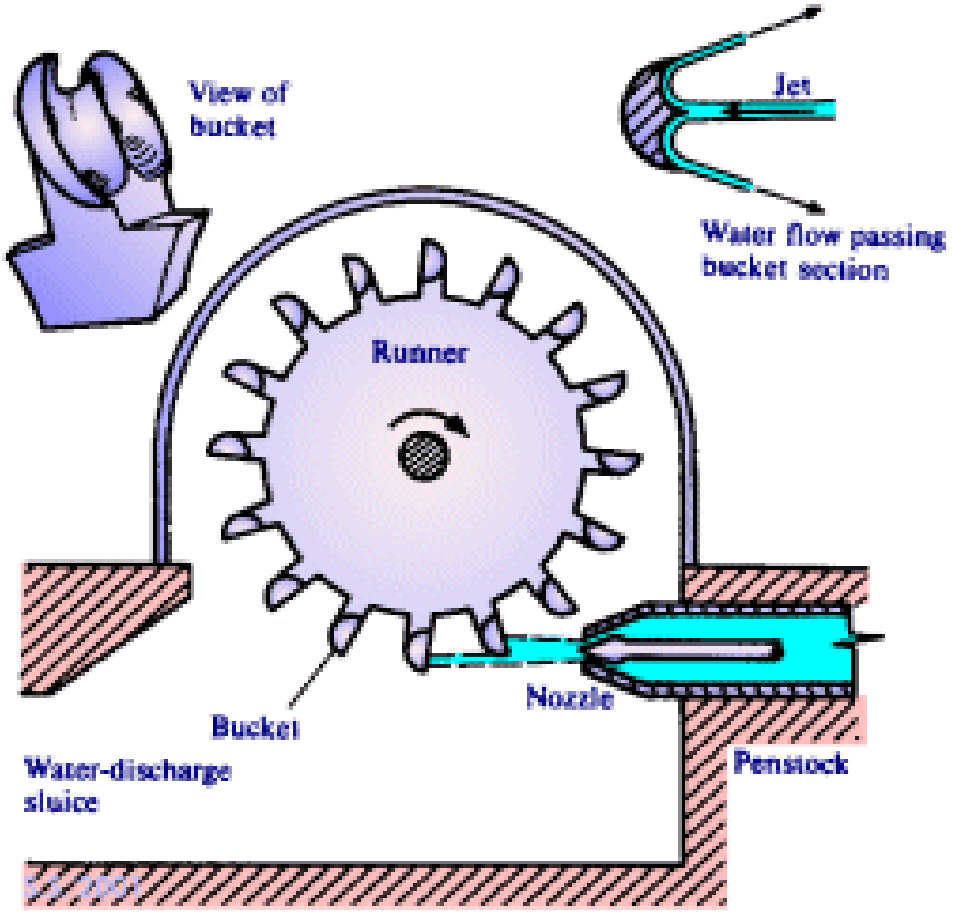
- In impulse turbine, the water's **potential energy is converted to kinetic energy** by nozzle and the jet of water is directed on the turbine's curved blades which reverse the flow.
- The resulting impulse spins the turbine and leaves the fluid flow with diminished kinetic energy.
- Newton's second law describes the transfer of energy for impulse turbine. No pressure change occurs at the turbine blades.

Pelton wheel

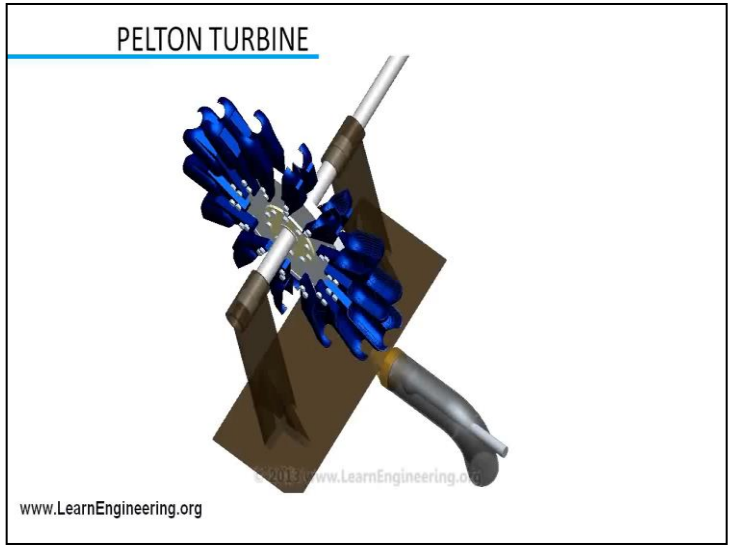
- Pelton wheel is a tangential flow impulse turbine, water flows along the tangent to the path of the runner.
- It operates under a high head of water and therefore requires a comparatively less quantity of water.
- Water is conveyed from the reservoir to the turbine through a ***penstock***. The penstock is connected to a branch pipe fitted with a nozzle. A powerful jet issue out of the nozzle, impinges on the buckets provided on the periphery of a wheel.

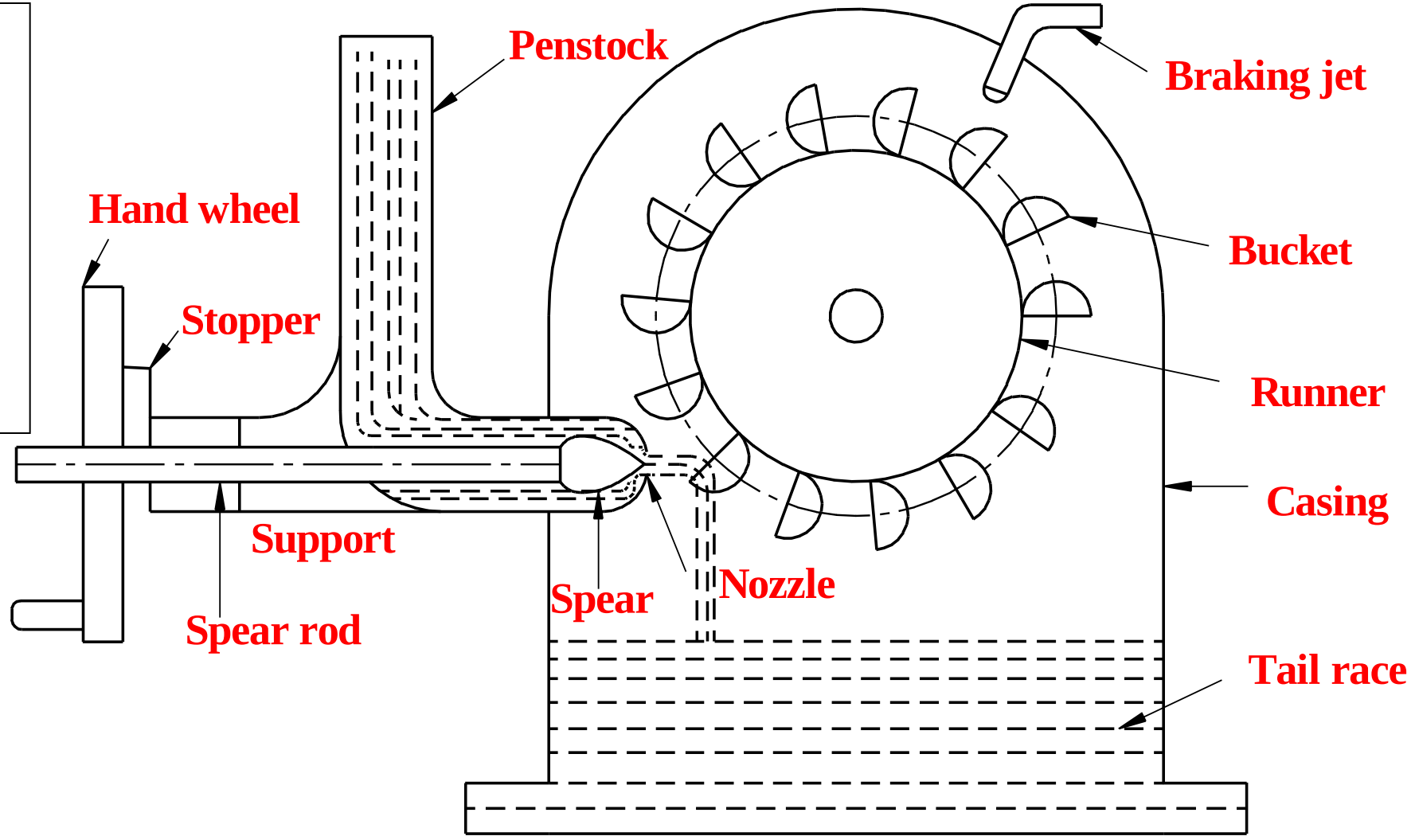
Pelton wheel





Pelton wheel





Single Jet Pelton Wheel

Pelton wheel

- The shape of the bucket is that of a double hemispherical cup having dividing wall known as splitter at the center. The splitter divides the impinging jet into two halves, which are deflected backward.
- As there is no pressure variation in flow , the fluid partly fills the buckets, and it remains in contact with the atmosphere. The nozzle is provided with a spear mechanism to control the quantity of water.
- The actual energy transfer from jet to wheel is by changing the momentum of the stream. The water after imparting its energy to the turbine is discharged into tail-race.

Pelton wheel

Advantages of Pelton wheel

- Simple in construction
- Easy maintenance

Disadvantage of Pelton wheel

- A lot of head loss occurs when the river discharge is low.

Reaction turbine (Parson's turbine)

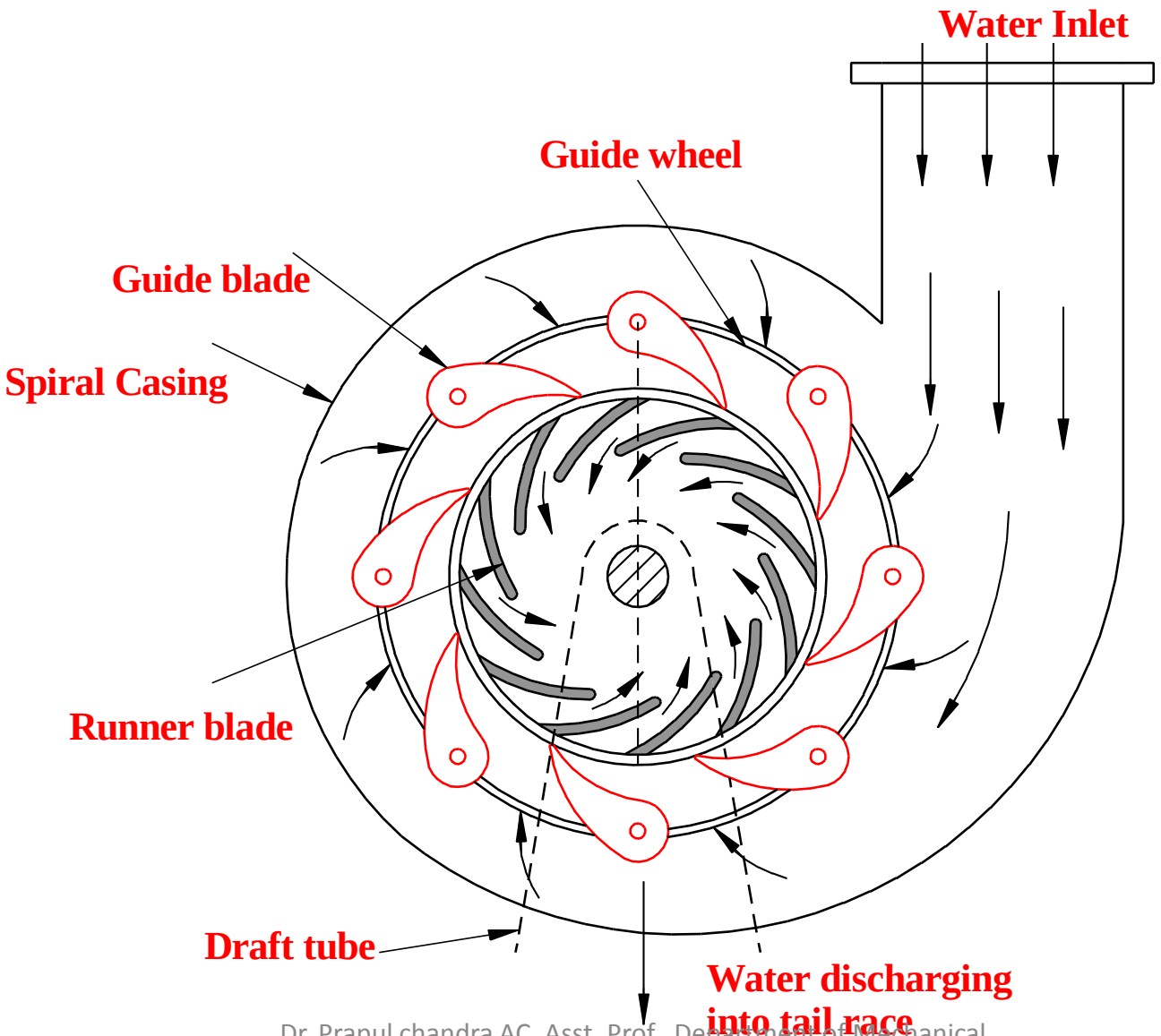
- ***Reaction turbines*** are acted on by water, which changes pressure as it moves through the turbine and gives up its energy.
- They must be enclosed in an air tight casing to contain the water pressure or they must be fully submerged in the water pressure or they must be fully submerged in the water flow.
- Newton's third law describes the transfer of energy for reaction turbines.

Reaction Turbine

The turbine is located between the high pressure water source and the low pressure water exit, usually at the base of the dam.

Most water turbine in use are reaction turbine. They are used in low and medium head applications, ***Francis turbine*** and ***Kaplan turbine*** are the examples of reaction turbine.

FRANCIS TURBINE- (Front View)



Francis turbine

- Francis turbine is a medium head inward flow reaction turbine.
- It consists of a spiral casing enclosing a number of stationary guide vanes fixed all round the circumference of the runner which carries moving vanes.
- Water at high pressure enters through the inlet and flows radially inwards through guide blades.
- The function of these guide vanes is to direct the fluid on to the runner at the required angle.

Francis turbine

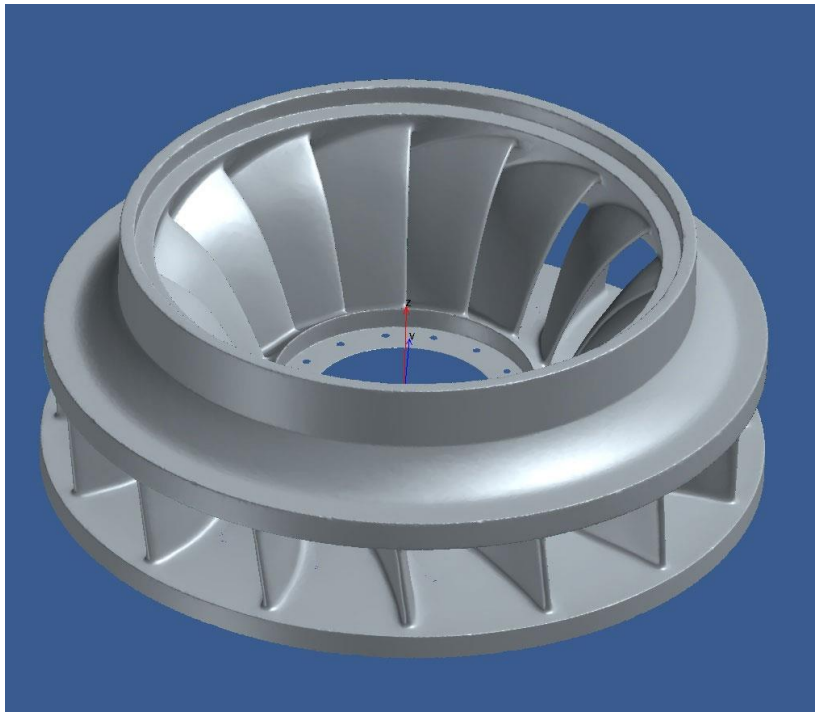
During its flow over the moving blades, the water imparts kinetic energy to the runner causing it to spin.

To enable the discharge of water at lower pressure, a diverging conical tube called ***draft tube*** is fitted at the center of the runner.

The other end of the draft tube is immersed in the discharging side of the turbine called '*tail race*'

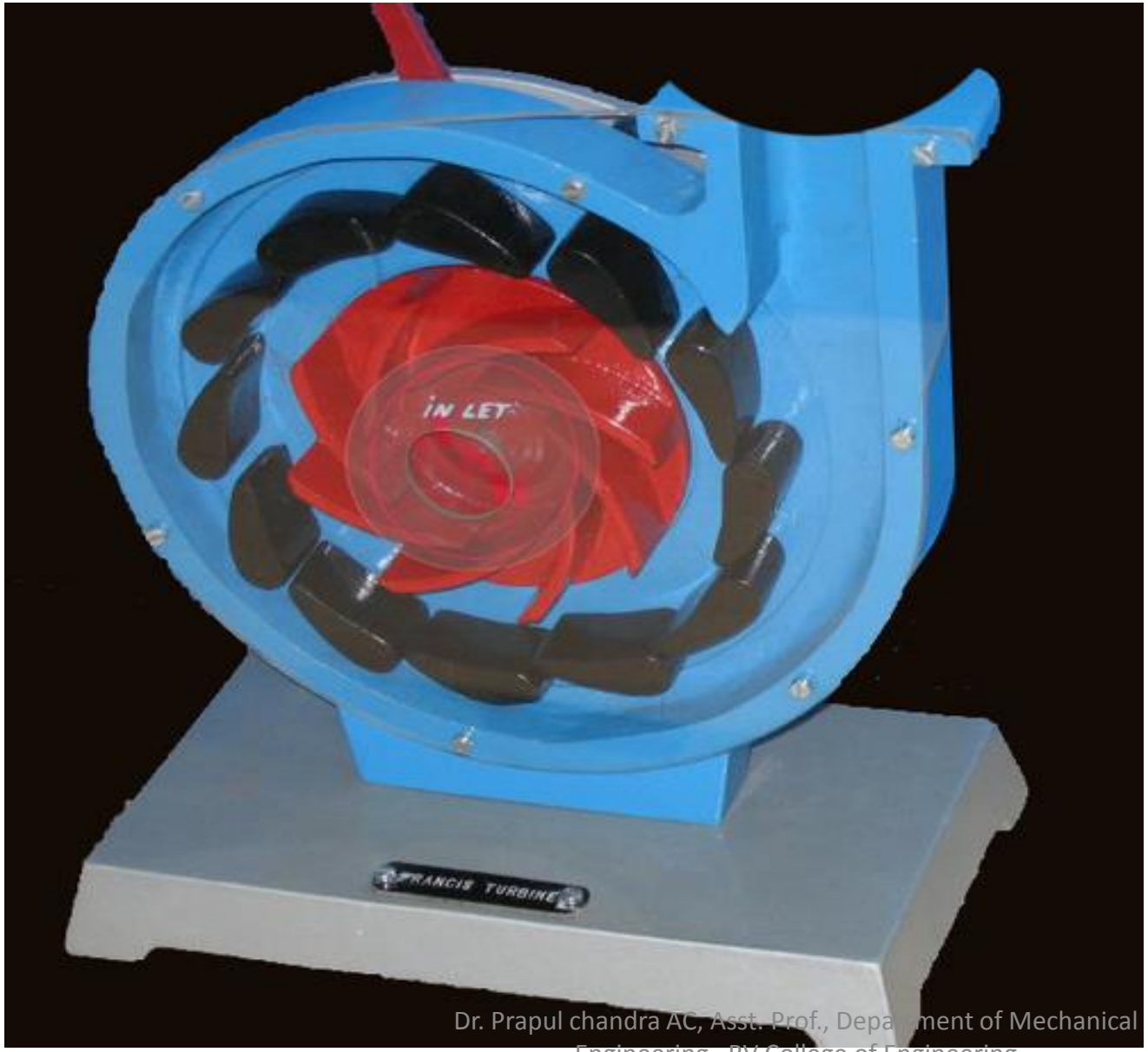
Francis turbine

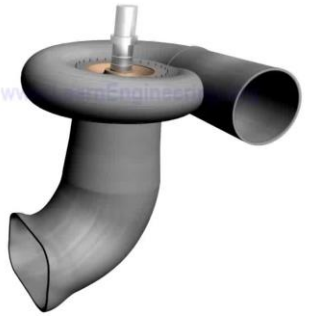




FRANCIS TURBINE

Francis turbine



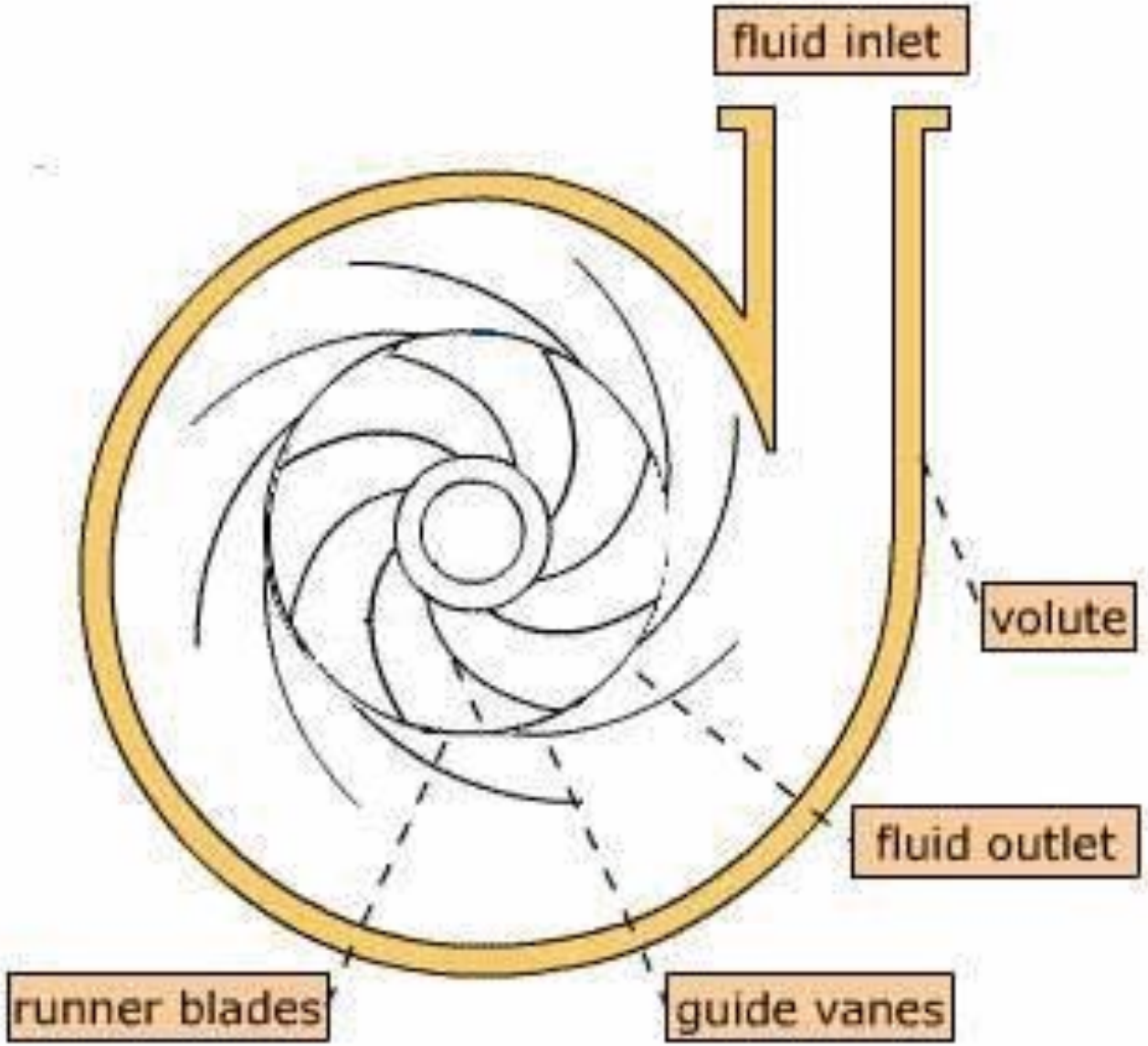


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Francis turbine

FRANCIS TURBINE

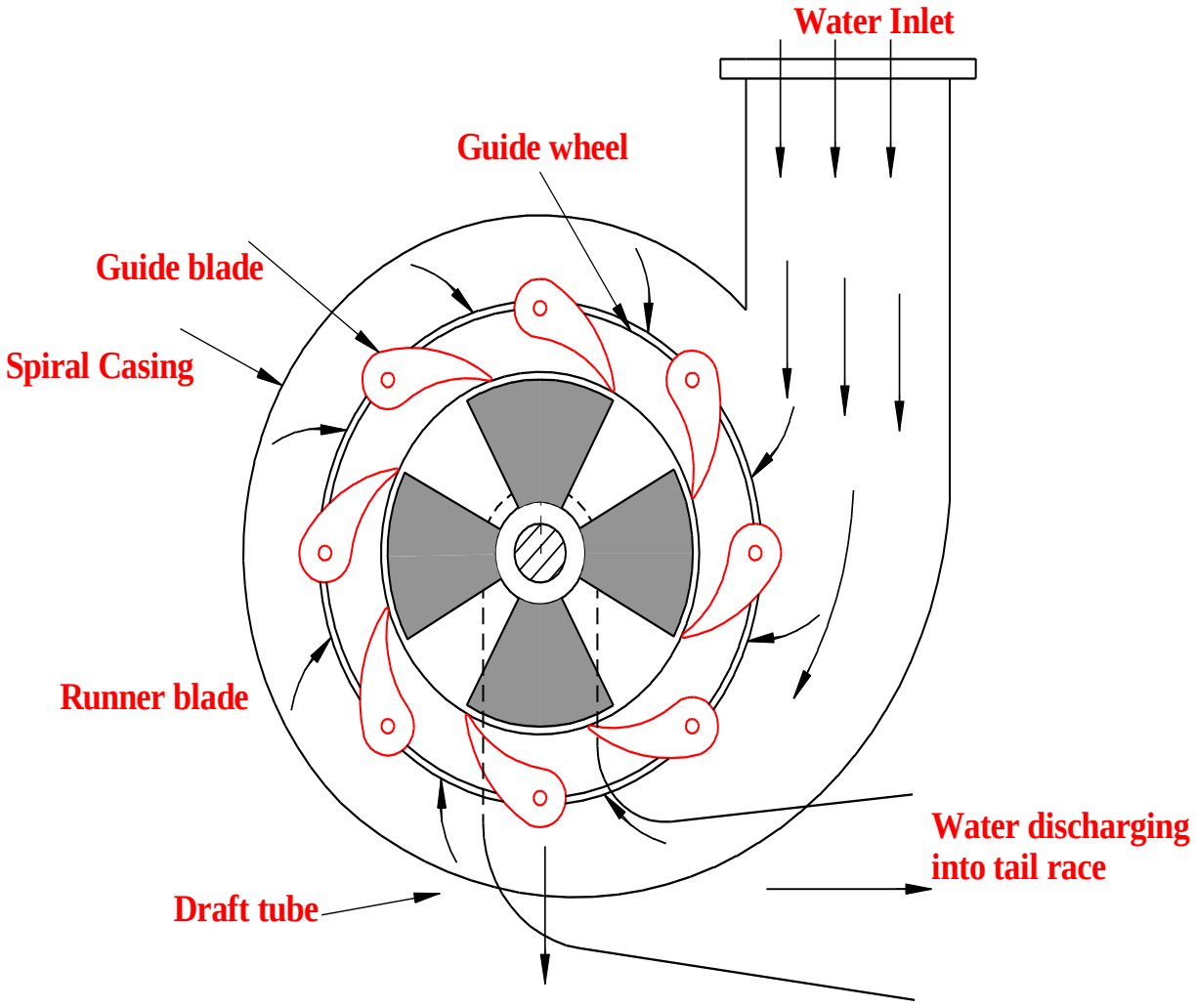
ANIMATION

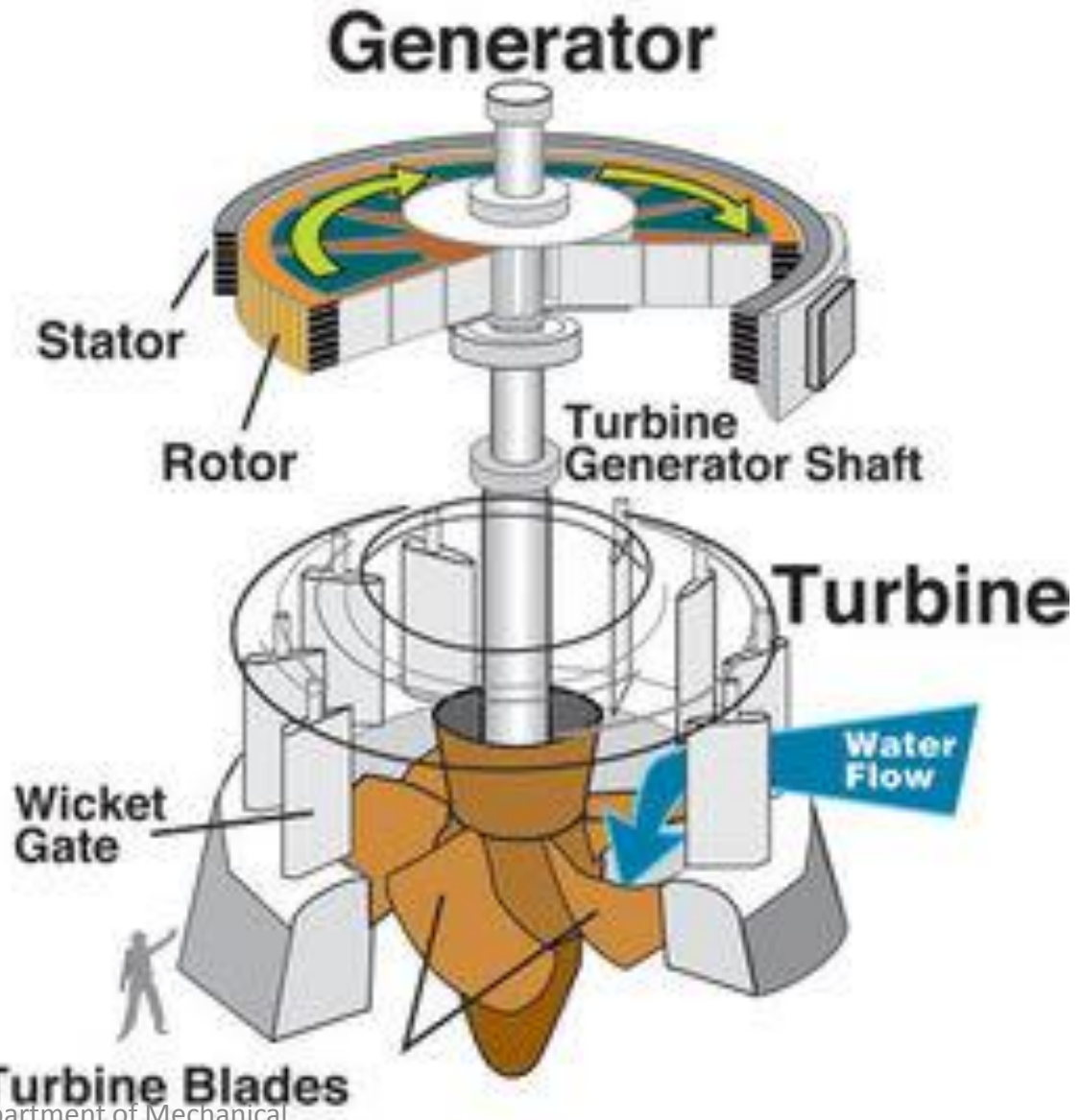


Kaplan turbine

- It is an axial flow turbine is used for low heads at high rotational speeds and large rates of flow.
- Kaplan turbine is an axial flow reaction turbine having small number of blades usually from four to six and closely resembles a ship's propeller.
- Water at high pressure enters the turbine casing through the inlet, flows over the guide vanes & then strikes the runner blades axially.
- The kinetic energy imparted to the runner blades set it into rotation.
- The water leaves at the center of the runner through a L shaped draft tube and is discharged into the tail race.

KAPLAN TURBINE (Top View)





KAPLAN TURBINE





KAPLAN TURBINE





Comparison of all three turbines

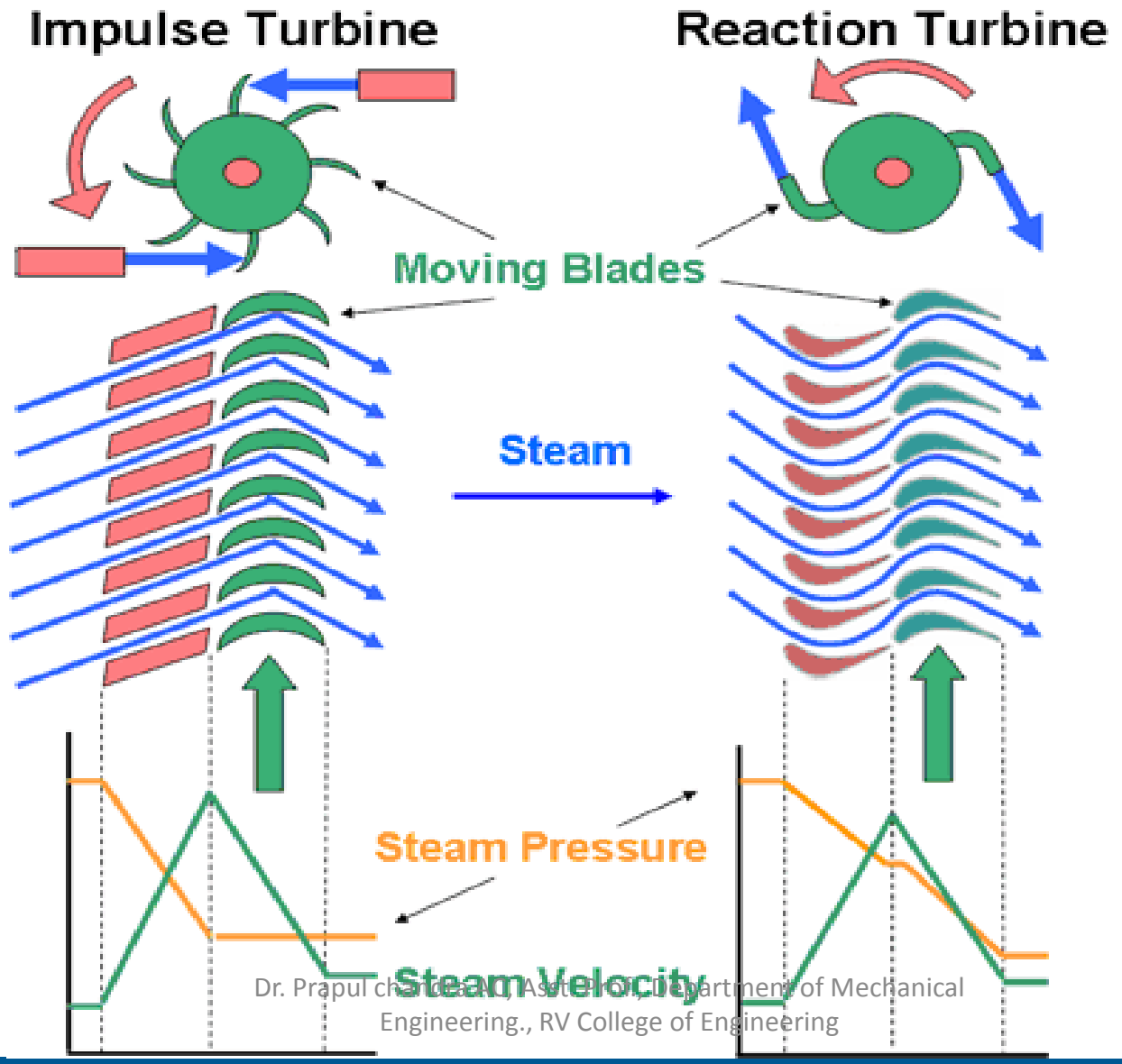


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Comparison between Impulse and Reaction Water turbines

Impulse Water Turbine	***Reaction Water turbines***
There is no pressure change of the fluid in the turbine rotor blades.	The pressure of the water changes as it passes through the turbine rotor blades
Impulse turbines do not require a pressure casement around the runner	A pressure casement is needed to contain the working fluid as it acts on the turbine
Draft tube is not necessary	Draft tube is necessary to discharge the water into tail race.
Newton' s second law describes the transfer of energy	Newton's third law describes the transfer of energy
Ex: Pelton Turbine	***Ex: Francis Turbine Kaplan Turbine***

Comparison between Impulse and Reaction Water turbines



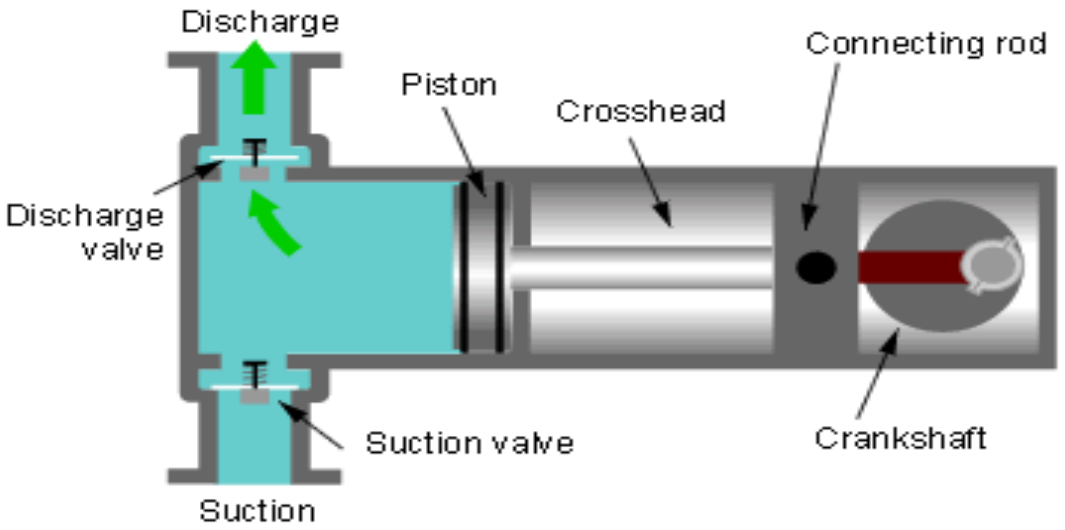
Gas Turbine Cycle

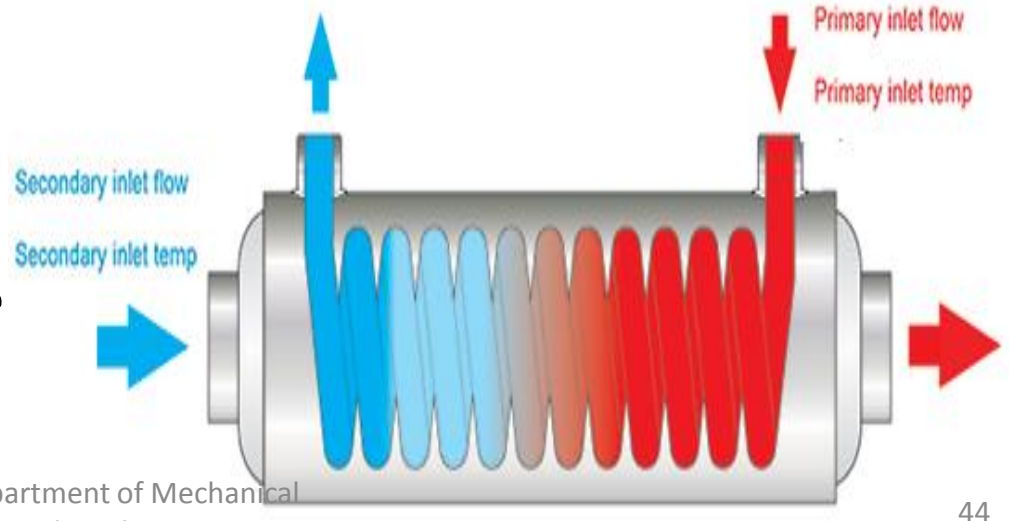
- Power generation, oil and gas, process plants, aviation.
- Gas Turbine is a prime mover. In this type turbine high pressure and temperature gas is passed on to the moving blades fixed to the runner.

- Gas Turbine cycle consists of:
  - Compressor
  - Combustion /heat exchanger
  - Turbine
  - Condenser (only in Closed Cycle)

Components of Gas Turbine

- **Compressor**
  - ✓Increases the fluid pressure by decreasing the volume.
- **Combustion Chamber**
  - ✓The air and fuel is passed in to a chamber. The air fuel mixture is then ignited which results in chemical reaction with heat release. Hence increasing the temperature.
- **Heat Exchanger**
  - ✓Equipment used to transfer heat from one fluid to another. The heat exchanger used to cool the liquid is called as condenser. In condenser the phase change may also occur

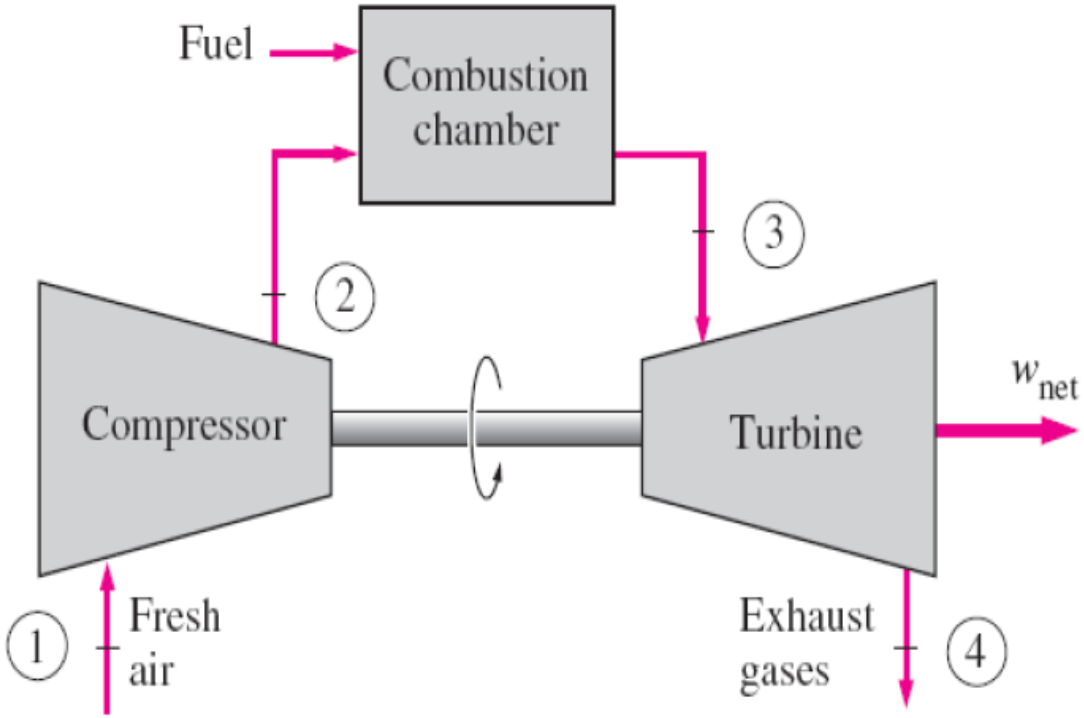




Open Gas Turbine Cycle

Principle:

The gas (air) with high pressure is passed in a chamber for combustion at constant pressure. The products with high temperature and pressure is passed in the turbine to expand and exhausted to atmosphere. The compressor is feed with fresh air. "*It is called open cycle because the exhaust gases does not enter the compressor. The compressor starts with fresh air.*"



Open Gas Turbine Cycle

Construction:

- This gas turbine consists of compressor, combustion chamber and the turbine. The turbine, compressor and generator are coupled coaxially.

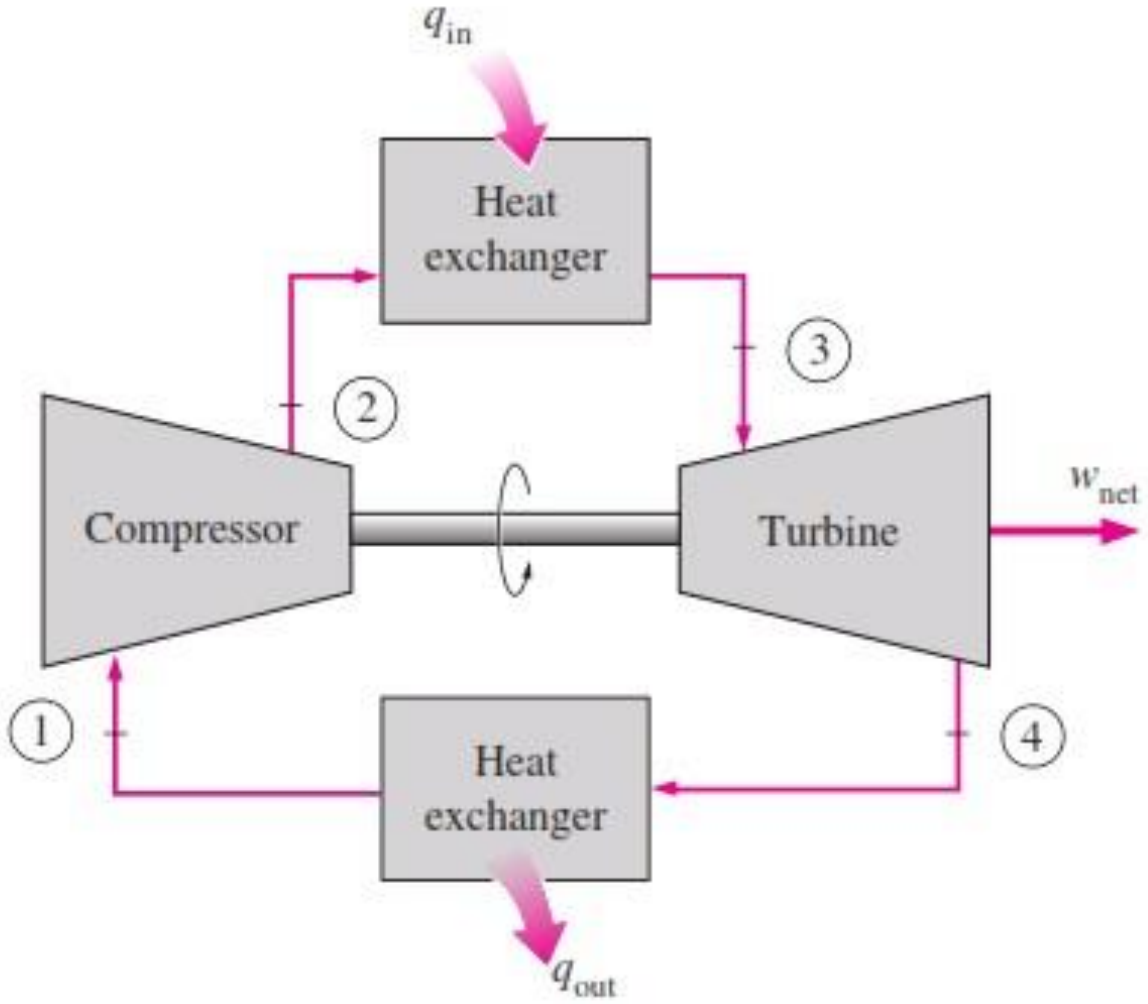
Working:

- The atmospheric air is drawn into the compressor to increase the pressure. The compressed air which is at high pressure is delivered to the combustion chamber and mean time fuel also being injected.
- Combustion occurs in the chamber at constant pressure. The products of combustion, which are at high temperature and pressure is made to flow in the turbine. There it expands and does the mechanical work.
- As a result the turbine and subsequently the generator. Since the compressor is coupled to the turbine, some mechanical work is spent in rotating the compressor

Closed Gas Turbine Cycle

Principle:

The gas (air) with high pressure is passed in a chamber for combustion at constant pressure. The products with high temperature and pressure is passed in the turbine to expand. **The exhausted gas is then passed back to the compressor through a condenser, where it is cooled and brought to original state.** *"Since, the same gas is circulating in the cycle hence it is called as closed gas turbine cycle"*



Closed Gas Turbine Cycle

Construction:

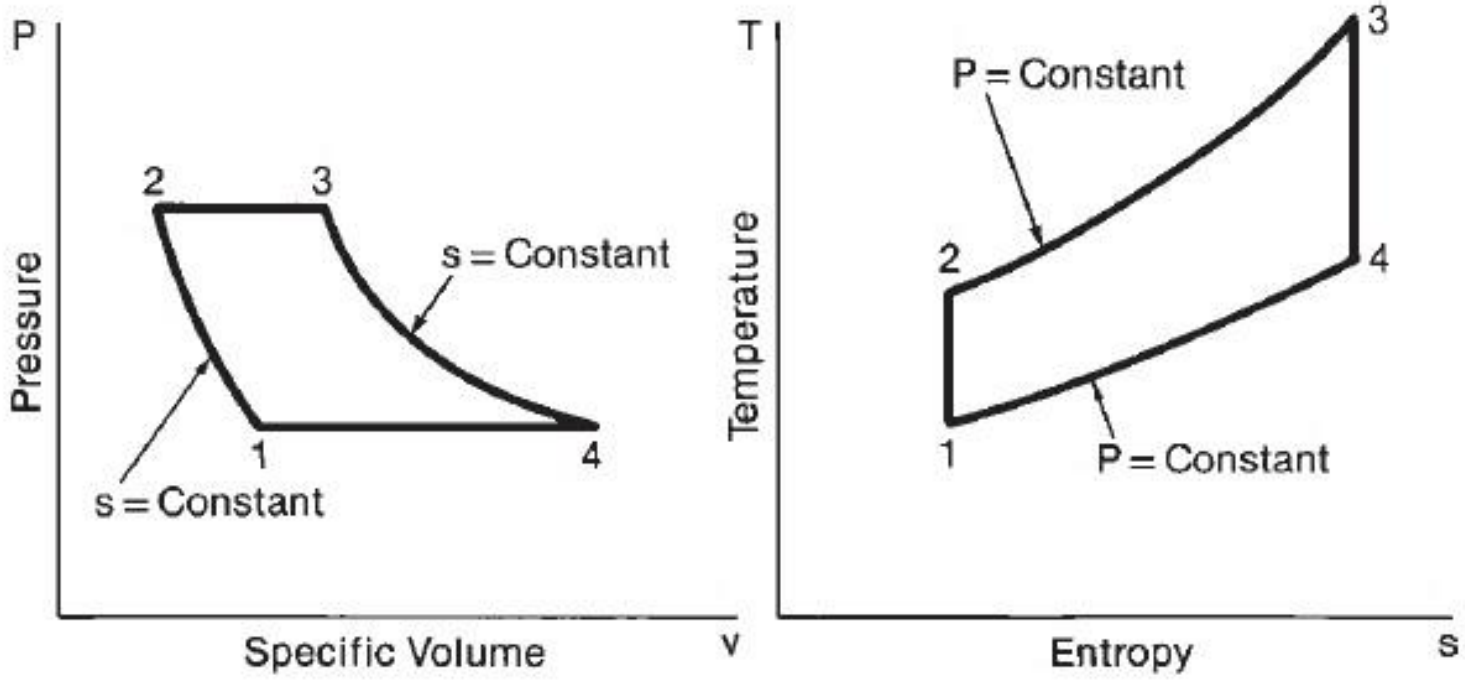
- This gas turbine consists of compressor, heat exchanger, turbine and condenser. The turbine, compressor and generator are coupled coaxially.

Working:

- The atmospheric air is drawn into the compressor to increase the pressure. The compressed air which is at high pressure is delivered to the heat exchanger
- The gas passes through the heat exchanger. Due to heat interactions there is a increase in gas temperature. The high temperature and pressure gas is made to flow in the turbine. There it expands and does the mechanical work.
- As a result the turbine and subsequently the generator rotates. Since the compressor is coupled to the turbine, some mechanical work is spent in rotating the compressor
- The gas the passes through a condenser where it is cooled to normal temperature and re-sent to compressor.

Brayton Cycle

It is the air standard cycle for gas turbine



- S – Entropy
  - Constant entropy is called isentropic process

- P- Pressure
  - Constant pressure is called isobaric process

Brayton Cycle

Isentropic process: A compressor takes in fresh ambient air (state 1), compresses it to a higher pressure leaving the compressor into combustion (state 2).

(the work transfers of the system are frictionless, and there is no transfer of heat or matter.)

Isobaric process: Fuel and the higher pressure air from compressor are sent to a combustion chamber (state 2), where fuel is burned at constant pressure. The resulting high temperature gases are sent to a turbine (state 3).

Isentropic process: The high temperature and pressure gases expand to the ambient pressure (state 4) in the turbine and produce power.

Isobaric process: The exhaust gases leave the turbine in to reservoir (open cycle) or cooled in heat exchanger (condenser) bringing back to its original state (state 1).

Advantages

Advantages over steam turbines

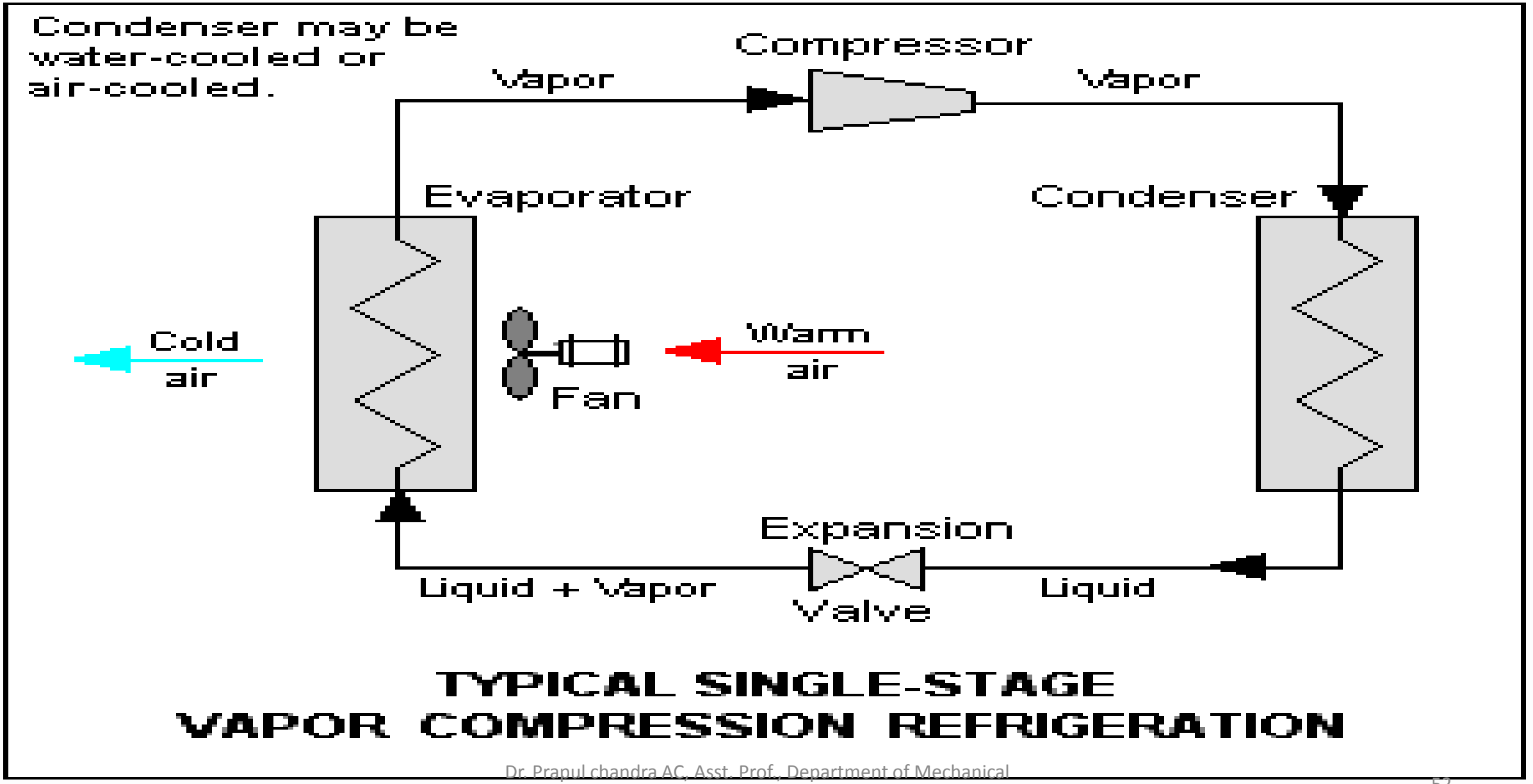
- Feed water and boiler are not required
- It is most reliable
- Its weight to power ratio is less
- Initial cost and maintenance costs are less
- Lubrication system is simple

General Electric Power-

Advantages over I.C Engines

- Mechanical Efficiency is higher
- Different fuels Can be used
- No vibration. Hence balancing problem is avoided.
- Less weight and more reliable
- Lubrication system is simple

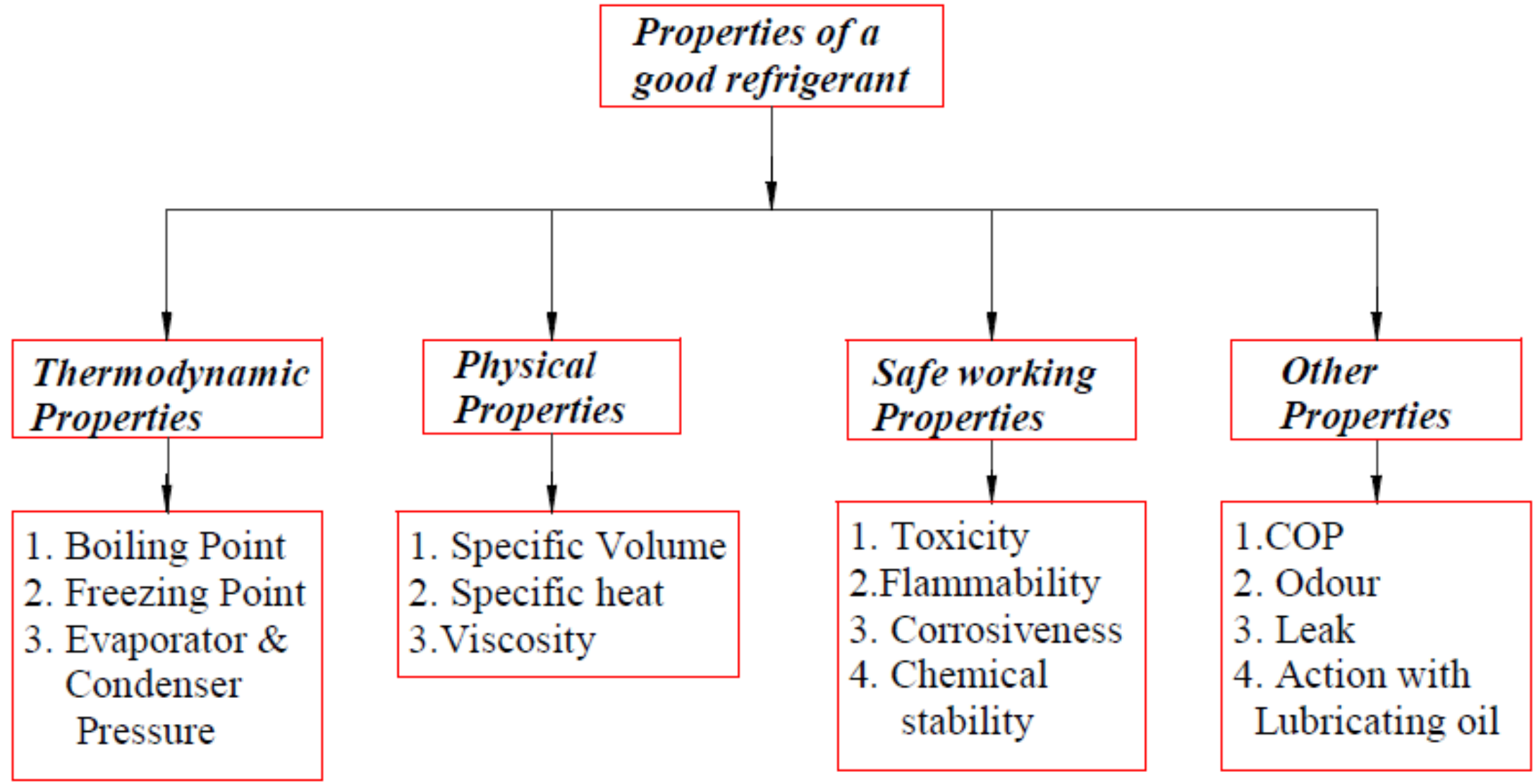
https://www.youtube.com/watch?v=zcWkEKNvqCA



TYPICAL SINGLE-STAGE VAPOR COMPRESSION REFRIGERATION

- Refrigeration is the removal of heat from a material or space, so that it's temperature is lower than that of it's surroundings.

- When refrigerant absorbs the unwanted heat, this raises the refrigerant's temperature ("Saturation Temperature") so that it changes from a liquid to a gas it evaporates. The system then uses condensation to release the heat and change the refrigerant back in to a liquid. This is called "Latent Heat".


Properties of a good refrigerant
Thermodynamic Properties
1. Boiling Point
2. Freezing Point
3. Evaporator & Condenser Pressure
Physical Properties
1. Specific Volume
2. Specific heat
3.Viscosity
Safe working Properties
1. Toxicity
2.Flammability
3. Corrosiveness
4. Chemical stability
Other Properties
1.COP
2. Odour
3. Leak
4. Action with Lubricating oil

DESIRABLE PROPERTIES OF A REFRIGERANT

Boiling point: An ideal refrigerant must have low boiling temperature at atmospheric pressure.

Freezing point: An ideal refrigerant must have a very low freezing point because the refrigerant should not freeze at low evaporator temperatures.

Evaporator and condenser pressure: In order to avoid the leakage of the atmospheric air and also to enable the detection of the leakage of the refrigerant, both the evaporator and condenser pressures should be slightly above the atmospheric pressure.

DESIRABLE PROPERTIES OF A REFRIGERANT

Latent heat of evaporation: The latent heat of evaporation must be very high so that a minimum amount of refrigerant will accomplish the desired result, in other words, it increases the refrigeration effect.

Specific volume: The specific volume of the refrigerant must be very low. The lower specific volume of the refrigerant at the suction of the compressor reduces the size of the compressor.

DESIRABLE PROPERTIES OF A REFRIGERANT

Specific heat of liquid and vapour: A refrigerant must have low specific heat when it is in liquid state and high specific heat when is vaporized. The low specific heat of the refrigerant helps in sub cooling of the liquid and high specific heat of the vapour helps in decreasing the super heating of the vapour. Both these desirable properties increase the refrigerating effect.

Viscosity: The viscosity of a refrigerant at both the liquid and vapour states must be very low as it is improves the heat transfer and reduces the pumping pressure.

Toxicity: The refrigerant should be non toxic, because any leakage of the toxic refrigerant increases suffocation and poisons the atmosphere.

DESIRABLE PROPERTIES OF A REFRIGERANT

Corrosiveness: The refrigerant should be noncorrosive to prevent the corrosion of the metallic parts of the refrigerators.

Chemical stability: An ideal refrigerant must not decompose under operating conditions.

Coefficient of performance: The coefficient of performance of a refrigerant must be high so that the energy spent in refrigeration will be less.

Odour: The refrigerant must be odourless, otherwise some food stuff such as meat, butter, etc. loses their taste. **Leakage tests:** The refrigerant must be such that any leakage can be detected by simple tests.

Action with lubricating oil: The refrigerant must not react with the lubricating oil used in lubricating the parts of the compressor.

LIST OF COMMONLY USED REFRIGERANTS

The most commonly used refrigerants are

- **Ammonia**–Used in vapour absorption refrigerator.
- **Carbon dioxide**–Used in marine refrigerators.
- **Sulphur dioxide**-Used in household refrigerators.
- **Methyl chloride**-Used in small-scale refrigeration and domestic refrigerators.
- **Freon -12** –Used in domestic vapour compression refrigerators.
- **Freon-22** –Used in air conditioners.

PRINCIPLE OF REFRIGERATION

In refrigeration, the heat is to be removed continuously from a system at a lower temperature and transfer it to the surroundings at a higher temperature.

- This operation according to the Second Law of Thermodynamics can only be performed by the aid of the external work.
- Therefore in a refrigerator, power is to be supplied to remove the heat continuously from the refrigerator cabinet to keep it cool at a temperature less than the atmospheric temperature.

REFRIGERATION EFFECT AND UNIT OF REFRIGERATION

In a refrigeration system, the rate at which the heat is absorbed in a cycle from the space to be cooled is called ***refrigerating effect.***

- The capacity of a refrigeration system is expressed in tons of refrigeration which is the unit of refrigeration.
- A ton of refrigeration is defined as the quantity of heat absorbed in order to form one ton of ice in 24 hours when the initial temperature of the water is 0^0c. Here it should be noted that one American ton (2000pounds) is taken as the standard in the refrigeration practice.

In S.I System
1 ton of refrigeration = 210 kJ/min
= 3.5 kW

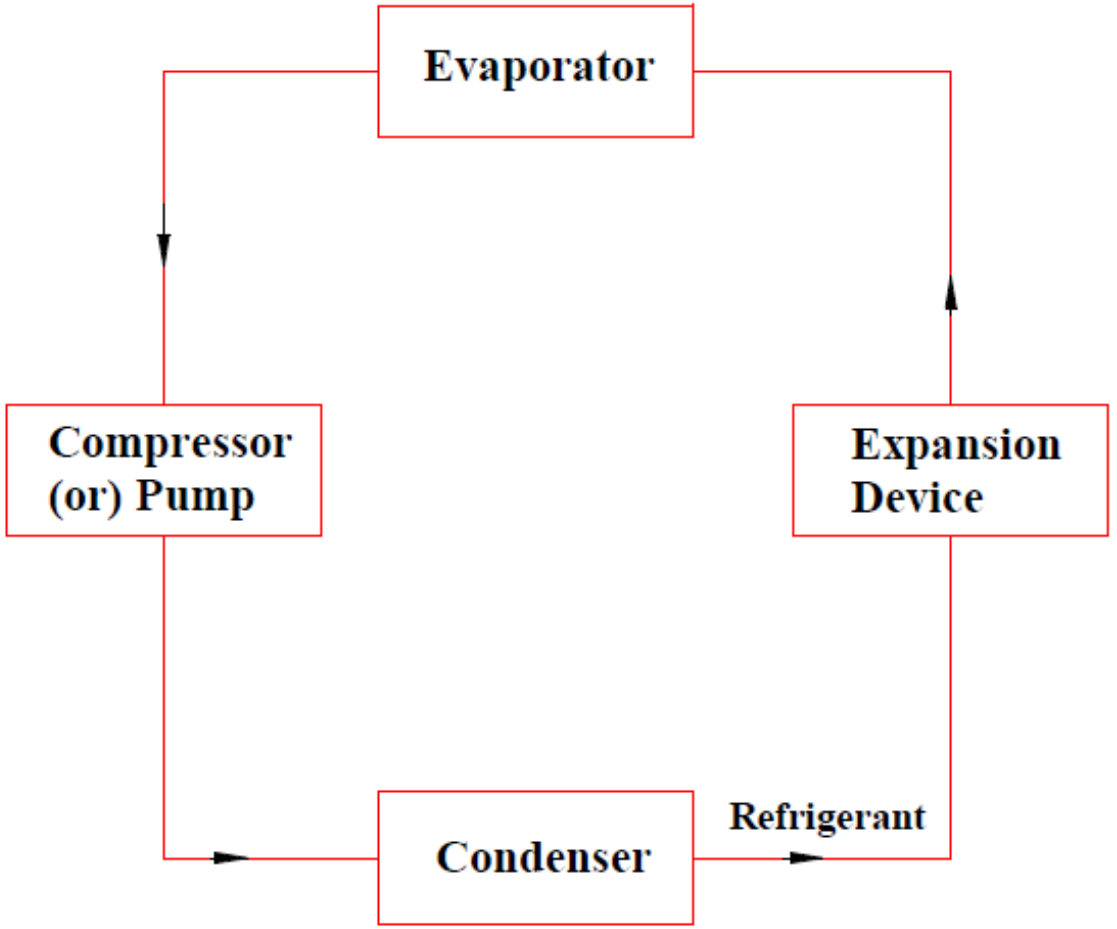
Coefficient of performance:

The performance of a refrigeration system is expressed by a factor known as the coefficient of performance (COP) . The COP of a refrigeration system is defined as the ratio of heat absorbed in a system to the work supplied. ***If Q=Heat Absorbed or Removed, kW***

W=Work supplied, kW

$COP = Q/W$

PARTS OF A REFRIGERATOR



PARTS OF A REFRIGERATOR

Evaporator: It consists of metal tubing which surrounds the freezing & cooling compartments to absorb heat and to evaporate the refrigerant.

Circulating system(Compressor or pump): It consists of mechanical devices powered by an electric motor which increase the pressure of the refrigerant and helps to circulate the refrigerant.

Condenser: Here, the latent heat is rejected from the refrigerant to atmosphere and it returns to liquid state and recirculated.

Expansion device: It is a device used to reduce the pressure and temperature of the liquid before it passes to the evaporator.

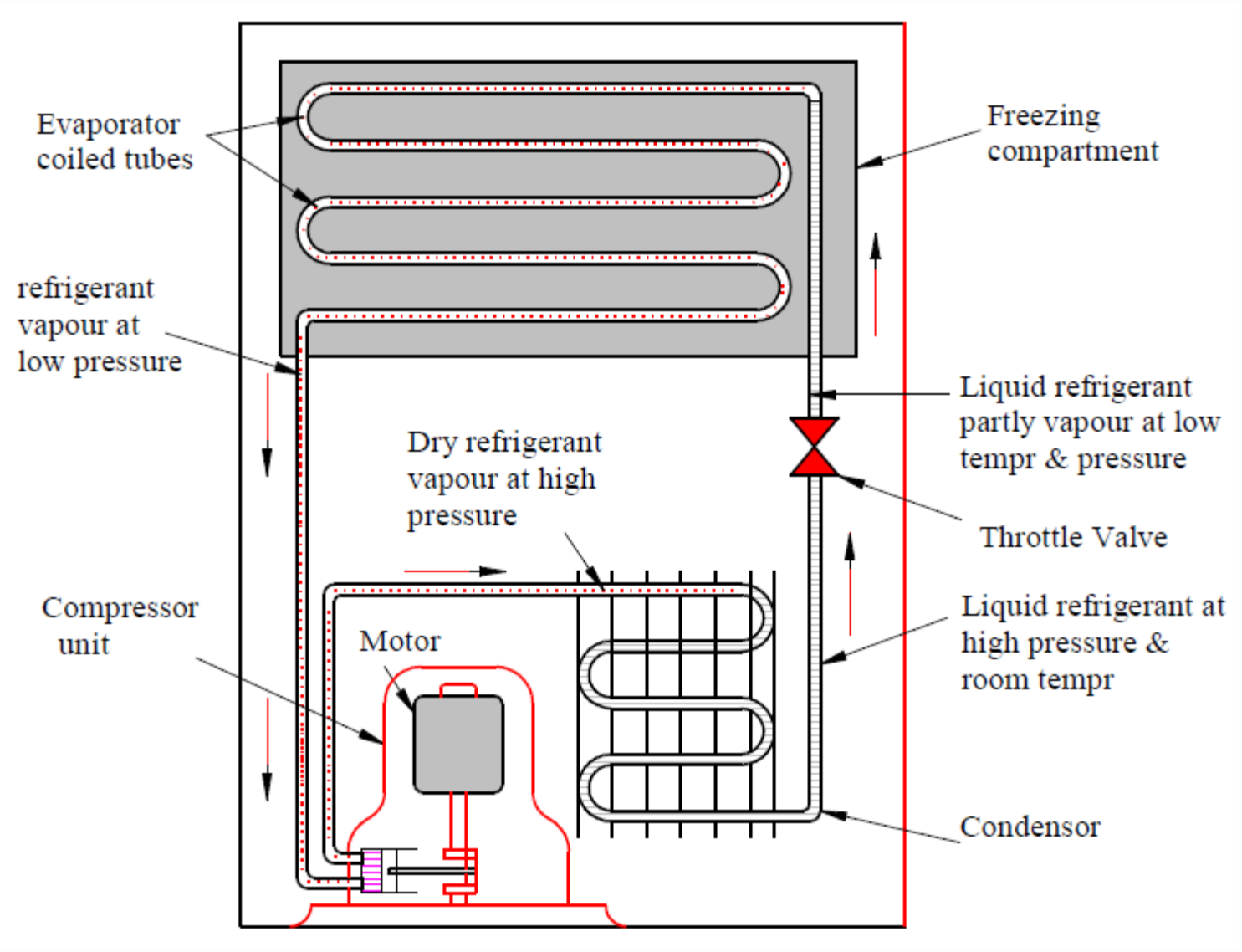
Types of Refrigeration systems

- ***Vapour Compression Refrigeration***
- ***Vapour Absorption refrigeration***

Vapour compression refrigeration makes use of mechanical energy supplied by the compressor to circulate the refrigerant.

Vapour absorption refrigeration makes use of the ability of a absorbent to absorb large volumes of vapour when cold & give off its vapours when heated.

VAPOUR COMPRESSION REFRIGERATION



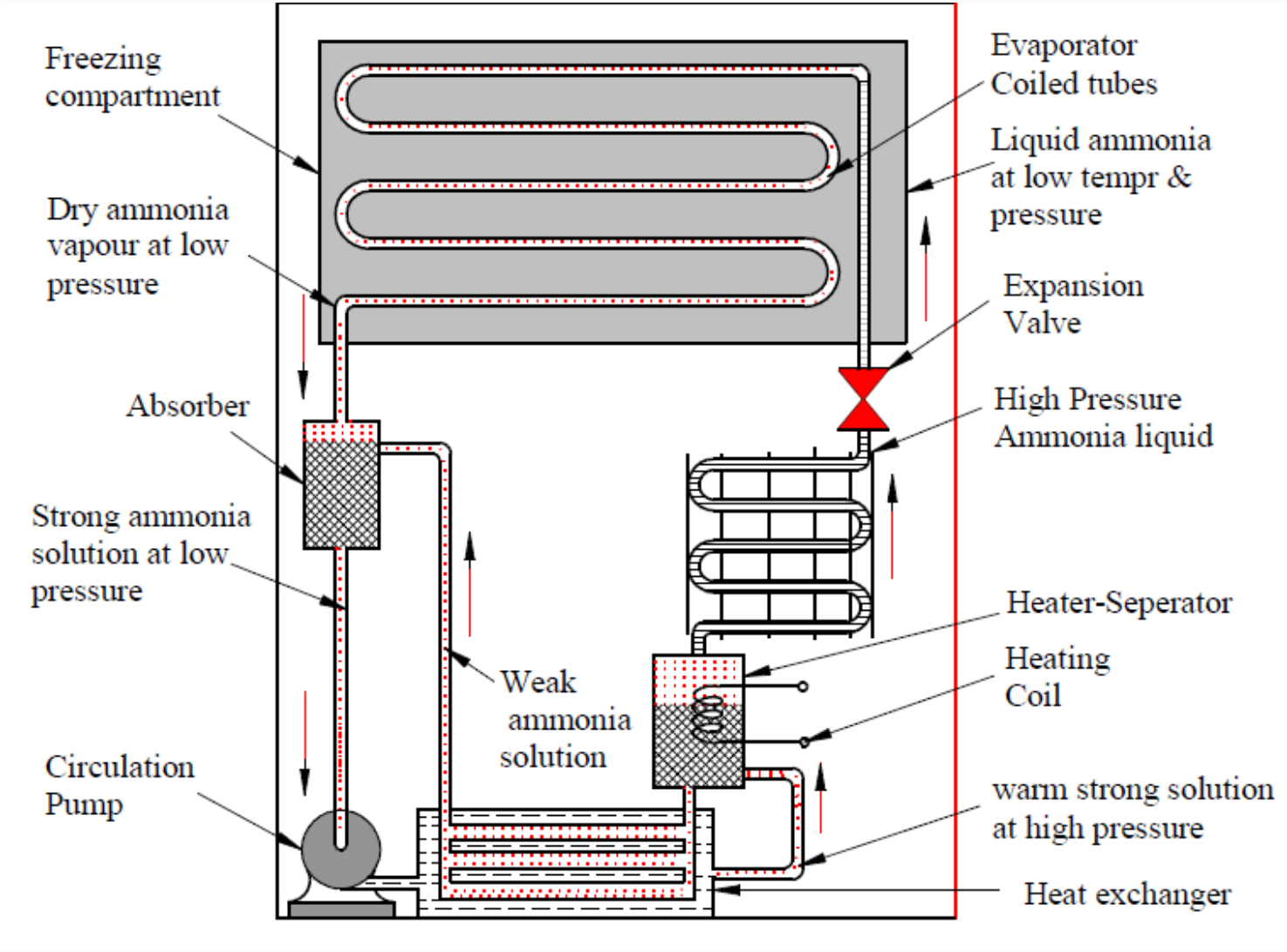
VAPOUR COMPRESSION REFRIGERATION

- The refrigerant in the form of vapour at low temperature & pressure enters the compressor and gets compressed to high temperature & pressure.
- The compressed vapours then enter the condenser where heat is transferred to a cooling medium such as air or water. The refrigerant is condensed to liquid form.

VAPOUR COMPRESSION REFRIGERATION

- The refrigerant then enters the expansion valve which reduces the pressure and temperature by throttling process. (toabout-100C)
- It finally passes on to the evaporator where it absorbs heat from surroundings and becomes low pressure dry vapour, and the cycle is repeated.

VAPOUR ABSORPTION REFRIGERATION



VAPOUR ABSORPTION REFRIGERATION

- The low pressure ammonia vapour leaving the evaporator is readily absorbed in the cold water in the absorber and becomes strong ammonia solution.
- The strong ammonia solution is pumped to the heat exchanger and the vapour gets separated due to high pressure & temperature.

VAPOUR ABSORPTION REFRIGERATION

- The ammonia vapours then pass to the condenser where it gets condensed in to liquid after rejecting heat to atmosphere.
- The high pressure ammonia liquid is then expanded in the throttle valve to get low pressure & temperature liquid.
- The low pressure & temperature ammonia liquid then enters the evaporator where it extracts the heat and gets vaporized. The cycle then repeats itself.

COMPARISON BETWEEN VAPOUR COMPRESSION & VAPOUR ABSORPTION REFRIGERATION

Vapour Compression	Vapour Absorption
Requires mechanical energy for compression of vapour.	Requires heat energy to separate the vapours from absorbent.
Maintenance is more	Less maintenance required.
The operation of the machine is noisy due to compressor	Quiet in operation due to absence of compressor
High operating cost due to requirement of electrical energy to the compressor.	Less operating cost due minimum consumption by the pump
Freon-12 is used as a refrigerant	Ammonia is used as refrigerant

Vapour Compression	Vapour Absorption
The design capacity is limited to 1000 tons	Design capacity can be above 1000 tons.
Chances of Leakage of refrigerant is more	No leakage of refrigerant
Charging of refrigerant is simple	Charging of refrigerant is difficult
Poor performance (COP) at part loads	System is not affected by variations in load